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Review

Measuring anxiety in zebrafish: A critical review

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ABSTRACT

Zebrafish are increasingly being used in behavioral neuroscience, neuropsychopharmacology and neurotoxicology. Recently, behavioral screens used to model anxiety in rodents were adapted to this species, and novel models which tap on zebrafish behavioral ecology have emerged. However, model building is an arduous task in experimental psychopathology, and a continuous effort to assess the validity of these measurements is being chased among some researchers. To consider a model as valid, it must possess face, predictive and/or construct validity. In this article, we first review some notions of validity, arguing that, at its limit, face and predictive validity reduce to construct validity. Then we review some procedures which have been used to study anxiety, fear or related processes in zebrafish, using the validity framework. We conclude that, although the predictive validity of some of these models is increasingly being met, there is still a long way in reaching the desired level of construct validity. The refinement of models is an ongoing activity, and behavioral validation and parametric research ought to advance that objective.

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Contents

1. Introduction 00
2. Behavioral models in neuropsychopharmacology: validity issues 00
3. A taxonomy of anxiety models in rodents and their equivalents in fish 00
   3.1. Learned behavior 00
      3.1.1. Pavlovian fear conditioning 00
      3.1.2. Avoidance conditioning 00
      3.1.3. Habituation of startle responses 00
   3.2. Innate behavior 00
      3.2.1. Response to predators 00
      3.2.2. Alarm reaction ("Schrecksreaktion") 00
      3.2.3. Open-field 00
      3.2.4. Novel tank diving test 00
      3.2.5. Scototaxis 00
      3.2.6. Inspection of a novel object 00
      3.2.7. Shoaling 00
   3.3. Analysis of neurovegetative responses in zebrafish 00
4. Conclusions 00
   Acknowledgments 00
   References 00

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1. Introduction

Zebrafish (*Danio rerio* Hamilton 1822), the cyprinid which became a central model in developmental biology and genetics [1], is increasingly being used for the study of neurobiology, neuropsychopharmacology and neurotoxicology [2–8]. As a model organism whose genome [9], transcriptome [10] and proteome [11] has been widely described, as well as being inexpensive and easy to maintain, the zebrafish has great potential as subject in high-throughput behavioral models in neuropsychopharmacology, neuropathology and psychopathology. In the last 10 years, the necessity of describing behavioral phenotypes which can be used as endpoints in this type of research is slowly being met. Among these, tests for anxiety, stress and fear are being developed [12–25]. As it happens with anxiety models in rodents, results with these tests are sometimes contradictory with each other. In this article, we will review some of these models, using as notional tool contemporary theories about model validity.

2. Behavioral models in neuropsychopharmacology: validity issues

In neuropsychopharmacology, as elsewhere in the behavioral neurosciences, the validity of behavioral models should always be taken with a grain of salt. In this regard, some criteria have been determined to analyze whether or not a given model is a "good" model for a given disease [26–34]. Influential works such as Willner's [26,34] and McKinney's [32] have provided the three criteria which are most widely cited in terms of validity: face, predictive and construct validity.

From a naïve standpoint, face validity is the degree of perceived resemblance between the model and its target. As van der Staay [33] argued, naïve approaches to face validity are always at risk of anthropomorphism, as researchers interpret certain behavioral displays as more or less alike symptoms of the target dysfunction. These naïve approaches were dismissed by both stricter and broader requirements; the first is that face validity, as defined by McKinney and Bunney [35], refers not only to nosology, but also to etiology, underlying processes and treatment; Holmes [36], for one, considered that the presence of a signal or symptom by itself has no heuristic value unless it is related to an underlying condition. A second, more influential approach is the requirement that face validity refers not to the full suite of symptoms which define a given pathology, but to the "core" symptoms—or, as it is being called nowadays, *endophenotypes* [28].

This approach is trivially applied in behavioral models of anxiety in mammals [27,37], in which fear, anxiety and stress are equated with species-specific defensive reactions. Threatening contexts elicit a set of species-specific behavioral, autonomic and neurophysiologic adjustments whose aberrant expression could represent a "fearful" endophenotype relevant to stress and anxiety disorders [38–43]. For example, one behavior that is expressed by many species at proximal levels of threat is the inhibition of all body movements and entrance into tonic immobility or freezing, which would prevent detection and attack by predators [41,43].

The defensive repertoire of rodents is a constellation of responses which include ultrasonic vocalizations, freezing, fleeing, "risk assessment", and defensive attack [40,41,43–45]. The particular set of responses that is emitted depends on the nature of the perceived threat [40,41,43,46,47]. Thus, risk assessment behaviors such as thigmotaxis, stretch-attend postures, and the differential distribution of exploration in "threatening" versus "safe" areas of open-fields, elevated plus-mazes and light/dark boxes appear in situations in which risk is merely potential; fleeing and freezing appear when the threat is distal; and freezing, fleeing and defensive attack are emitted when risk is proximal [41,43,46]. From the point of view of face validity, models which elicit risk assessment are better models of anxiety, while models which elicit freezing, fleeing and defensive attack are better models of panic [48].

Table 1
Most commonly used behavioral models of anxiety and fear in rodents, along with common measures of avoidance, activity and arousal.

Model	Measure
Elevated plus-maze	*Avoidance*: Time spent in open arms *Activity*: Time spent in closed arms; total entries *Arousal*: Blood pressure; heart rate; number of fecal boli; number of rearings
Open-field	*Avoidance*: Time spent in the center field; thigmotaxis *Activity*: Total distance travelled *Arousal*: Blood pressure; heart rate; number of fecal boli; number of rearings
Light–dark box	*Avoidance*: Time spent in light area *Activity*: Activity in dark area; total entries *Arousal*: Blood pressure; heart rate; number of fecal boli; number of rearings
Inhibitory avoidance	*Avoidance*: Latency to enter conditionally aversive environment *Activity*: No intra-test measure *Arousal*: Blood pressure; heart rate; number of fecal boli; number of rearings
Contextual fear conditioning	*Avoidance*: Magnitude of freezing behavior *Activity*: No intra-test measure *Arousal*: Blood pressure; heart rate; number of fecal boli; number of rearings

The face validity of models of anxiety relies heavily on behavioral validation [38]. Toth and Zupan [49] argued that an analogy can be made between a "fearful" endophenotype and "emotionality" in non-human animals, a multidimensional trait that is composed of measures of *av*oidance of novel environments, *ac*tivity/behavioral inhibition in highly or moderately threatening situations, and autonomic *ar*ousal (AvAcAr). These measures can be obtained using both conditioned and unconditioned models of anxiety.

Table 1 presents some common measures of avoidance, activity and arousal in the three most common unconditioned exploration models (elevated plus-maze, open-field, and light–dark box) and two conditioned fear models (inhibitory avoidance and contextual fear conditioning) in rodents. Notice that the same rationale underlies exploration models: avoidance of novel environments is measured by the time spent in a threatening area (open arms/center field/light area), while an independent measure of activity can be recorded as time and activity in a non-threatening novel area (closed arms/peripheral field/dark area). Of equal notice is that measures of arousal are usually the same in both conditioned and unconditioned models of fear and anxiety, with behavioral measures (vertical activity or rearing and defecation) appearing more commonly. Also notice that, even though fear conditioning and avoidance tasks have the advantage of allowing precise parametric manipulations of stimulation parameters such as shock duration and intensity [50], no internal, independent measure of activity has been described. Purely anxiogenic manipulations should increase avoidance of threatening situations and not have an effect on activity, while producing magnitude-dependent increases in arousal, and purely anxiolytic manipulations should have the opposite result [49].

It should be noticed that, in order to understand face validity in a more broad conceptual network, we had to appeal to construct

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validity [51], and the specific topography of the behavioral display (that is, the specific behaviors which are being analyzed) had to be switched with the general functional significance of behaviors across species [41,43,51–53].

Most theories regarding model validation simply require that predictive validity is met [26,30,32,34,35,54,55]. "*Predictive validity* of a model is broadly defined as the ability to make accurate predictions about the human phenomenon of interest based on the performance of the model" ([55], p. 446). It stands to reason that failure to predict future findings in the target disease should invalidate a model; this requires, again, that we embed predictive validity in the broader context of construct validity [51,52]. Pharmacological isomorphism is predictive validity in a stricter sense [56]. Pharmacological isomorphism refers to the ability that a given model possesses in the identification of drugs with potential therapeutic value in humans.

Construct validity refers to the "theoretical soundness" of animal models. "The construct validity is not established by determining the relation between a test and an accepted criterion but is instead based on the establishment of relationships, which are in turn based on the definition of a trait. Implicitly, a construct is defined by a network of associations" ([33], p. 126). That amounts to saying that the model is not (and should not be) developed in a theoretical vacuum, and embodies a construct (e.g., anxiety, or the AvAcAr approach of Toth and Zupan) that is, in its turn, part of an expansive theory. We now turn to these models *per se*, in an attempt to judge their validity in accordance with what has been described.

3. A taxonomy of anxiety models in rodents and their equivalents in fish

Some authors [39–41,43,48,53,57–59] have proposed that a differentiation between fear, anxiety and panic can be made on the basis of "defensive distance", a psychological construct which describes the perceived immediacy of threat and risk. Risk that is merely potential should elicit risk assessment-like behavior and a specific spatiotemporal distribution of exploratory and other behavior that is equivalent to a state of anxiety; distal risk elicits escape and avoidance, thus being equated with fear; and proximal risk elicits urgent active (defensive attack, escape) or passive (freezing, tonic immobility) behavior. Frederico Graeff proposed that each of these levels are processed and controlled by different structures in a rostrocaudal defense circuitry [58].

Among taxonomies of anxiety models in mammals, the most widely used are those which differentiate between learned and innate behavior [30,38,48,60–63]. This approach is usually taken because of pragmatic questions: it is far easier to implement an innate exploratory behavior model of anxiety such as the elevated plus-maze or the open-field than to establish a baseline response using conditioning. However, there are marked differences in the neural underpinnings of, for example, fear conditioning and fear elicited by predator exposure [64]—differences which should be important when considering the translation of these models to zebrafish. We will follow this basic distinction here, mainly to connect with the wider literature on anxiety (Table 2).

3.1. Learned behavior

There are two basic conditioning protocols applied to the study of fear in animals: Pavlovian conditioning and operant escape/avoidance conditioning. In Pavlovian (or classic) conditioning, the control that a given eliciting stimulus (the unconditioned stimulus, or US) over a given unconditioned response (UR) is transferred to other stimuli (the conditioned stimuli, or CS) that do not initially elicit the UR [65]. In the Pavlovian conditioning of fear, the US is usually a noxious or aversive stimulus, such as electric shock, air puffs, or others. In operant conditioning, the response of the animal is contingent to a consequence, which will reinforce future instances of that response under the same environmental context [66]. Operant conditioning of escape responses target responses which "turn off" aversive stimuli (e.g., flight, bar pressing), while operant conditioning of avoidance target responses which allows the organism to circumvent aversive stimuli which were previously associated with an environmental context or with signaling stimuli; thus, in these contingencies, responding is increased by both negative and positive reinforcers as punishment is avoided [67–71].

3.1.1. Pavlovian fear conditioning

The Pavlovian conditioning of many different reflex systems, under a wide array of conditions, has been reported in fish, which show the typical processes of overshadowing, blocking, autoshaping, and higher order conditioning [69–73]. Pavlovian conditioned bradycardia and eye retraction responses in response to a shock-signaling conditioned stimulus was studied in goldfish [74–76], but few descriptions of classical conditioning were made using zebrafish. An interesting exception is the work by Hall and Suboski [77] on conditioning of *Schrecksreaktion*, the species-typical response to alarm substances (described below). These authors paired an olfactory stimulus (morpholine, the CS) with an alarm substance derived from the injured skin of three conspecifics (the US); this pairing transferred control of typical responses associated with the *Schrecksreaktion* to the conditioned olfactory stimulus. The authors also paired a neutral visual stimulus (a red light-emitting diode) with the alarm substance, obtaining similar results.

These experiments are interesting, since they compare to findings in rodents. Exposure to the odor of predators (cats) to rodents produce contextual fear conditioning after a single trial, with subjects presenting freezing to the context in which they were exposed to the partial predator stimulus [78–80]. Interestingly, the benzodiazepine midazolam exerts an anxiolytic effect to the first exposure to cat odor, but not to the second exposure [80], a phenomenon that resembles "one-trial tolerance" in the elevated plus-maze [81].

The principal shortcoming of fear conditioning approaches is that they might be models of fear, not anxiety. Fear-potentiated startle, for example, lacks pharmacological isomorphism as a model of anxiety, as serotonergic drugs do not block this effect [82]. In the case of visual or olfactory conditioning to alarm substances in zebrafish, no pharmacological studies were reported so far and therefore we cannot judge the predictive validity of this model. To show that this phenomenon is analogous to cue or context conditioning of antipredator responses to cat odor in rodents, however, pharmacological isomorphism is still insufficient, as other requirements (e.g., establishment of contextual fear conditioning, "one-trial tolerance" to the effects of benzodiazepines) are necessary for predictive validity.

3.1.2. Avoidance conditioning

Avoidance conditioning is usually divided in two types: inhibitory (passive or one-way) and active (or two-way) avoidance. In this operant task, an aversive consequence is contingent to either a signal or a context, and the conditioned behavior avoids presentation of this aversive consequence [69]. Inhibitory avoidance associates a given signaling stimulus with aversive conditions, while associating another stimulus ("S^{Δ}") with "safety" conditions. Thus, two processes are present in the acquisition of avoidance: the reinforcing effect of terminating or avoiding stimuli that have

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Table 2
Q5 Models used to study anxiety and fear in zebrafish analyzed in this article, along with common avoidance, activity and arousal measures.

Model	Protocol reference	Measure
Conditioned behavior		
Conditioned alarm reaction	[77]	*Avoidance*: Bottom-dwelling *Activity*: Total path length *Arousal*: ? *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming
Shuttle box avoidance	[12]	*Avoidance*: Percent of avoidance responses *Activity*: No intra-test measure *Arousal*: ? *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming after signal
Inhibitory avoidance	[87]	*Avoidance*: Latency to enter conditionally aversive environment *Activity*: No intra-test measure *Arousal*: ? *Other*: No other intra-test measures
Habituation of fast-start	[122]	*Avoidance*: Number of elicited fast-start responses *Activity*: No intra-test measure *Arousal*: ? *Other*: No other intra-test measures
Unconditioned behavior		
Response to predators/predator inspection	[15,233]	*Avoidance*: Distance from predator *Activity*: No intra-test measure *Arousal*: ? *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming, bottom-dwelling
Fast-start readiness	[117]	*Avoidance*: Number of elicited fast-start responses *Activity*: No intra-test measure *Arousal*: ? *Other*: No other intra-test measures
Open-field	[130]	*Avoidance*: Time spent in the center field; thigmotaxis *Activity*: Total distance travelled *Arousal*: ? *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming, bottom-dwelling
Novel object inspection	[203]	*Avoidance*: Time spent near object; distance from object *Activity*: No intra-test measure *Arousal*: ? *Other*: No other measures
Novel tank diving test	[132]	*Avoidance*: Time spent in bottom of tank; number and latency of transitions to upper half *Activity*: Total path length *Arousal*: ? *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming
Scototaxis	[24]	*Avoidance*: Time spent in white compartment *Activity*: Shuttle frequency *Arousal*: Ventilatory frequency *Other*: Burst swimming, thigmotaxis, freezing, erratic swimming
Schrecksreaktion	[171]	*Avoidance*: Bottom-dwelling; place avoidance *Activity*: Total path length *Arousal*: ? *Other*: Burst swimming, erratic swimming, freezing
Shoaling	[136]	*Avoidance*: Average distance among shoal mates *Activity*: Total path length (focal fish) *Arousal*: ? *Other*: No other measures reported

been previously paired with aversive stimuli, and the production of stimuli ("safety signals") that have been paired with the absence of shock [69–71,83–86].

In the zebrafish, both inhibitory and active avoidance conditioning have been established. In the first case, a robust response was established after a single trial in which the animal had to refrain from swimming from a white to a dark compartment in order to avoid receiving an electric shock [87]. If the NMDA-receptor antagonist MK-801 is administered between the acquisition and recall trials (that is, during consolidation) in a dose that does not impair swimming [88], the consolidation of this aversive memory is blocked. NMDA receptors are important for the establishment of long-term potentiation in the zebrafish telencephalon [89]. These results are consistent with results in rodents, in which inhibitory avoidance conditioning is also dependent on NMDA receptors in the forebrain [90]; they are also consistent with amnesic effects of MK-801 in avoidance acquisition and Pavlovian fear conditioning in goldfish [91,92].

Interestingly, sleep deprivation in zebrafish impairs retention or retrieval of aversive memories in inhibitory avoidance in zebrafish [93]. Sleep-deprived fish spend more time in the portion of the tank in which punishment (mild electric shock) happened during 5-min sessions for 5 consecutive days prior to 24-h sleep deprivation. These effects seem to be a general effect on learning and/or memory, since preference for a positively reinforced environment also decreases after the same sleep deprivation and training schedule [93]. These results are consistent with data on sleep deprivation, anxiety and learning in humans, whose responses to deprivation

are mainly related to learning, with little (if any) worsening of anxiety [94].

The telencephalon of teleost fishes has been suggested as an homologue of the limbic system of mammals [95], and there is at least some evidence that the dorsomedial telencephalon (Dm) is functionally equivalent to the amygdala of mammals [96,97]. In goldfish, Dm ablation impairs the acquisition of two-way (active) avoidance, while lesions in the dorsolateral telencephalon do not [96]. Since zebrafish and goldfish present extensive similarities in the organization of their brains [98], we should expect that lesions of Dm in zebrafish also produce the same effect. This work still awaits technical developments in zebrafish neuroscience.

Another important system which has been investigated in the zebrafish is the neural nicotinic receptors. Nicotine facilitates the acquisition [99] and retention [100] of avoidance in mice; the administration of the nicotinic receptor antagonists mecamylamine and dihydro-β-erythroidine impairs short and long-term memory and retrieval of a one-trial inhibitory avoidance in rats [101]. Likewise, nicotine improved performance in an aversive spatial alternation task in which the fish was punished for entering a compartment it had entered in the previous trial [102,103]. Nicotine also increases the levels of dihydroxyphenylacetic acid (DOPAC), the primary dopamine metabolite, in this task [104]. The role of dopamine was also observed in inhibitory avoidance in goldfish [105], which suggests a more general role of nicotine and dopamine on long-term memory [106] than on fear *per se*.

Apart from neurotransmitter systems, cell adhesion molecules were also implicated in avoidance conditioning in zebrafish—although the procedures used were of the active type, and not inhibitory avoidance. In a first experiment [107], animals had to cross a hurdle to avoid mild electric shocks after a light signal. One hour after this task was finished, the authors injected monoclonal antibodies against the HNK-1 epitope or polyclonal antibodies against ependymins intracerebrally; those animals showed decreases recall of the task after 2 days [107]. In a second experiment, zebrafish which were trained to avoid mild electric shocks after a light signal showed increased expression of L1.1 and NCAM mRNA in the optic tectum than untrained fish and non-learners; after that, monoclonal antibodies for L1.1 were injected intracerebrally 1 h after acquisition of the task, and animals were exposed to the apparatus 2 days later. Those animals presented less signs of recalling the task than did non-injected animals [108]. Those results are interesting, since L1.1 and NCAM were implicated in anxiety in knockout mice [109,110]. It has also been suggested that cell adhesion molecules are necessary for memory consolidation [111].

So far, it seems that avoidance conditioning in zebrafish is similar, in terms of pharmacological and neural bases, to the same task as applied in rodents. However, it also suffers from the same shortcomings that avoidance conditioning presents in other species. For one, most drugs and manipulations which affect avoidance conditioning also affect learning of other operant contingencies; thus, it is virtually impossible to pharmacologically dissociate learning and emotional effects of these manipulations on the acquisition and maintenance of avoidance memories. As noted by Susan Mineka, the main problem with avoidance conditioning is that, even though we can clearly determine that fear is a major determinant in acquisition, the source of reinforcement for the maintenance of responding is increasingly more obscure as conditioning proceeds because each successful avoidance is also a Pavlovian extinction trial [112]. Thus, fear diminishes over the course of a run of consecutive avoidance responses, but the avoidance response itself does not become weaker [112], which diminishes the construct validity of this task. It also follows that drugs which alter the acquisition process manipulate fear, and thus performance during acquisition should be a better index of these states than consolidation or retrieval [113]. However, since some drugs which are thought not to affect fear also impair avoidance acquisition, the passive avoidance task lacks predictive validity.

3.1.3. Habituation of startle responses

Fish present a fast-start response to noxious or "startling" stimuli that is defined as a high-energy "burst" swimming which dislodges the animal away from the eliciting stimulus [114]—which can be visual (predators or predator models), mechanossensory (sounds or objects dropped in the water) or nociceptive (shocks) [115]. The response itself is classified as a fixed action pattern, triggered by the firing of a pair of giant brainstem neurons called Mauthner cells [114,116–119]. Though being a fixed action pattern, the fast-start is also liable to modulation by the context in which it is elicited, and is not solely controlled by the rhombencephalic Mauthner cells; in goldfish and Siamese fighting-fish, for example, telencephalon ablation decreases the frequency of startle responses to stimuli, and discrete lesions to the dorsomedial telencephalon accelerate the habituation rate of fast-starts [120,121]. The habituation rate also seems to be mediated by glutamatergic and cholinergic mechanisms; Best and colleagues [122] described a steep habituation curve for the fast-start response of zebrafish larvae (7 days post fertilization) to acoustic stimuli, an effect which is mediated by NMDA glutamatergic and nicotinic cholinergic receptors. In zebrafish, fast-start responses are observed with great frequency during the exposition to alarm substances [22], which could be an analogue of fear-potentiated startle.

The *space cadet* (*spc*) locomotion mutant shows interesting alterations in fast-start swimming. These mutants (which show a defect in axon pathfinding of a subset of commissural axons that control locomotor behavior) display abnormalities in fast-start responses, showing spontaneous fast-starts while swimming [123,124]. Again, further studies are necessary to assess the utility of this mutant in anxiety research.

So far, the actual extension of contextual modulation of the startle response by fear-eliciting stimuli has not been evaluated. The previous exposition to an alarm substance shows promise, as it increases the readiness of fast-starts, but it has not yet been tested whether or not alarm substances increase sensitivity to fast-start or decrease habituation rate. In any circumstance, it is dubious that fast-start responses can be used as a measure of anxiety, as they probably are recruited by fear-related stimuli. Exaggerated startle responses are characteristics of patients with post-traumatic stress disorder, and not generalized anxiety disorder [125]. Since Mauthner cells receive GABAergic and serotonergic input [119], it would be rather difficult to determine whether the effects of systemic administration of benzodiazepines or serotonergic drugs are due to a decrease in anxiety or to an effect on these inputs. Thus, at present, fast-start habituation is a poor model for anxiety or fear responses in fish, lacking predictive and construct validity.

One attempt at testing the effects of drugs on fast-start readiness, without prior habituation, was made using ethanol in three strains of zebrafish (wild-type, long-fin striped, and blue long-fin). Ethanol inhibited fast-starts in both wild-type and long-fin striped zebrafish, though a smaller dose was needed for inhibition in the latter than in the former [126]; at the higher concentrations used (1.0%, v/v), ethanol did not produce any effects on startle responses of blue long-fin zebrafish. Interestingly, long-fin zebrafish seem to be more anxious than wild-type zebrafish as assessed by the novel tank diving test [18] (Table 3), but this effect was borderline significant.

3.2. Innate behavior

While conditioning allows the careful establishment of stimulus control relationships, allowing the investigator to readily identify

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Table 3
Strain differences in zebrafish tests of anxiety.

Strain	Behavior	Reference
Long-fin wild-type	Leftwards shift of dose-response to alcohol in fast-start readiness, in relation to wild-type	[151]
AB	Leftwards shift of dose-response to alcohol in novel tank diving test, in relation to Leopard, short-fin wild-type and long-fin wild-type Decrease startle, in relation to WIK No anxiolytic effect of nicotine Increased freezing in the central compartment of the modified scototaxis test, in relation to GloFish or PETCO	[122,126,182]
WIK	No bottom-dwelling	[182]
PETCO	No anxiolytic effect of nicotine	[182]
Albino	Increased bottom-dwelling, in relation to wild-type	[18,25]
Leopard	Increased bottom-dwelling, in relation to wild-type	[18,25]
TM1	Increased predator inspection, in relation to SH and Nadia	[138]
SH	Decreased shoaling tendency, in relation to TM1 and Nadia	[138]
Nadia	Increased latency to feed after disturbance, in relation to TM1 and SH	[138]

the parameters which induce fear or anxiety in its animals, they have the disadvantage of requiring food deprivation or delivery of shock, as well as obliging an extensive period of pre-training to shape the desired response. In the case of unconditioned models, on the other hand, training is not needed, but "without extensive analysis, it is not clear what the reinforcers are—whether, for example, any particular behavior is driven by positive [...] or negative [...] drives" [51]. They show, however, increased validity, as they present "dubious" or "diffuse" threatening stimuli and are more close to species-specific patterns of anxiety [127]. In zebrafish, as in rodents, most of these models use the exploratory behavior of the animal in a highly aversive novel environment (center of an open-field, top of a novel tank, light area of a dark/light tank, region nearest to a predator) as an index of anxiety, controlling for locomotor variability by assessing the exploratory behavior in a mildly aversive novel environment (periphery of an open-field, bottom of a novel tank, dark area of a dark/light tank, region farthest to a predator) or total locomotion.

A trend in the investigation of anxiety using exploratory behavior in rodents is the analysis of drug effects on ethogram measures—that is, on specific, discrete behavioral acts which happen when the animal is exploring the apparatus [38,128]. In zebrafish, this approach is still incipient, but shows great promise. Freezing, thigmotaxis, fast-starts, leaping out of the tank, bottom-dwelling, hyperactivity, shoaling and erratic swimming were proposed as indices of anxiety [15,19,22,129–138] (Table 4). Among those, freezing, thigmotaxis and erratic swimming are more commonly analyzed across models.

Freezing is usually defined as complete or partial immobility, with the animal performing only opercular beats, oculomotor responses and those fin movements needed to maintain a dorsal posture; this measure is increased in the white compartment of the scototaxis test [24], and is highly predictive of increased anxiety in this test [17]. Repeated confinement to the white compartment decreases freezing without decreasing the time spent in the white compartment in the scototaxis test [139]. In the novel tank diving test, freezing is increased after exposition to an alarm substance [18] (but see [22]), but not by hypoxanthine-3-*N*-oxide (a synthetic substance thought to represent the "minimum chemical structure" of the alarm substance; see below) [140] or the presentation of predators or predator models [15,132]. In the novel tank diving test (see below), freezing shows intra- and inter-session habituation [141], not seen in the scototaxis test [142,143]. The pharmacology of freezing is rather straightforward; the benzodiazepine receptor agonist diazepam decreases freezing in the novel tank diving test [144], while caffeine has no effects [18,25]. Thus, from a pharmacological and behavioral point of view, at least in zebrafish, freezing seems to be related to anxiety but not to fear [17]. This contrasts with freezing in rodents, which is defined as the complete absence of body movements in response to unavoidable fear stimuli [145].

Thigmotaxis refers to a "centrophobic" behavior in which animals spend the majority of the time of a session near the walls of the apparatus. It was first reported in the open-field in guppies [146] and latter in the same test in zebrafish [19,129,134,147,148], and remains the primary measure of anxiety in this model. This behavior is sensitive to ethanol [134,149] and to the benzodiazepine receptor antagonist FG-7142, as well as to cocaine withdrawal [19,148]. In the scototaxis test, thigmotaxis in the white compartment does not show intra- or inter-session habituation, but the proportion of the trial that the animals spend in thigmotaxis in the white compartment decreases after being thrice confined ("flooding") to the white compartment [139]. However, in the same test animals which show high avoidance of the white compartment show increased thigmotaxis while confined to the *black* compartment, though [17]; Blaser and colleagues interpreted these results in light of the "affordances" theory of defensive behavior [150], stating that the walls of the black compartment may afford the opportunity for escape, while the walls of the white compartment do not, inducing freezing instead. In any case, in the dark/white preference test, "it appears that thigmotaxis does not yield the most sensitive measure, but further research will be needed to better understand what this behavior signifies in this and other tasks" [17].

Erratic swimming refers to a highly unpredictable pattern of "zig-zagging" swim path in which zebrafish continues to move but the locomotion is interrupted by small direction changes [18,22,25,130,141,144]. This pattern of swimming can be measured either by sampling turn angles in different time bins of a session (e.g. [17,129,130,151]) or using more sophisticated measures from dynamical analysis; the Richardson dimension, an estimate of the degree with which a path departs from a straight line [152–154], presents a high degree of reliability to measure erratic movements in goldfish [155] and should be useful in assessing erratic swimming in models of anxiety in zebrafish. Erratic swimming is not affected by exposure to a sympatric predator in zebrafish [15], but is affected by exposure to a predator model [129,132] or exposure to an alarm substance and hypoxantine-3-*N*-oxide [18,22,25,140]. In the scototaxis test, no differences between high-avoidant and low-avoidant fish confined to either the white or black compartment were observed in erratic swimming, but this behavior loaded on the "white avoidance" factor when it was scored in the white compartment [17]. In the novel tank diving test or in the scototaxis test, erratic swimming shows no intra- or inter-session habituation [139,141]. Erratic swimming, then, seems to be a response to fear-inducing stimuli. Consistent with these observations, a sub-convulsive dose of pentylenetetrazole impairs habituation of erratic swimming in the novel tank diving test [141], an effect

Table 4
Three behavioral endpoints used in different unconditioned models of anxiety in zebrafish.

Endpoint	Observations	References
Freezing: Full immobility, with the exception of opercular and oculomotor movements.	Increased frequency in the white compartment of the scototaxis test Increased in high-avoidance fish during confinement to the white compartment of the scototaxis test Habituates in the novel tank diving test, but not in the scototaxis test Sensitive to alarm substance Insensitive to predators and H_3NO Sensitive to benzodiazepines and ethanol Insensitive to caffeine	[17,18,24,25,141–144]
Thigmotaxis: More time spent in the periphery than in the center of the apparatus	Main measure of anxiety in the open-field Does not habituate in the scototaxis test Decreases after "flooding" in the scototaxis test Increased in high-avoidance fish during confinement to the white compartment of the scototaxis test Sensitive to benzodiazepine antagonists and ethanol	[17,19,129,134,147–149]
Erratic swimming: Unpredictable, fast changes in swimming direction.	Sensitive to predator models, alarm substance and H_3NO Insensitive to sympatric or allopatric predators Loads in "white avoidance" factor in scototaxis No habituation in novel tank diving test Pentylenetetrazole and caffeine impair habituation in novel tank diving test Sensitive to chronic fluoxetine treatment Insensitive to chronic diazepam treatment Sensitive to diazepam and morphine withdrawal	[15,17,18,22,25,132,140,141,144]

which is also observed with caffeine (although caffeine does not increase erratic swimming overall [18,25]). Similarly, chronic fluoxetine treatment dramatically decreases total erratic swimming in the novel tank diving test [18,25] but does not affect habituation rate [141]. In addition, chronic treatment with diazepam does not decrease erratic swimming in the novel tank diving test, while diazepam withdrawal greatly increases this measure [144]—as does morphine, but not ethanol or caffeine, withdrawal [144].

3.2.1. Response to predators

The response of animals to predators has been used as a model of anxiety [45,156]. In general, animals present different responses to threatening stimuli (including predators), depending on the "defensive distance" of the source of threat; distal risk elicits escape and avoidance responses, proximal risk elicits freezing, tonic immobility and defensive attack, and potential risk elicits risk assessment [40,43,48]. The exposure to a predator, then, should elicit behaviors controlled by proximal defensive distances—and, therefore, behaviors classified as fear-like [64,157].

Bass and Gerlai [15] demonstrated that laboratory-reared zebrafish are capable of discriminating sympatric and allopatric predators. In relation to a no-stimulus control or to allopatric predators and sympatric and allopatric harmless fish, the sympatric predator produced an increased number of jumps, but no effect on other measures of anxiety (fast-starts, erratic movement, freezing); these results are not surprising, since the mean distance from the stimulus was not different across groups, which indicate that neither allopatric nor sympatric predators produced avoidance. When AB zebrafish were presented with an animated computer image of the same sympatric predator (the Indian leaf fish *Nandus nandus*), an increase in the frequency and duration of jumping and erratic swimming was observed [132]. Zebrafish also presented increased cortisol concentrations after direct or visual contact with a predator [14], which suggest that exposure to predators, predator models or partial predator stimuli induce fear rather than anxiety.

A confound may be observed in these experiments, as they tested animals in groups rather than individually. In zebrafish, as well as in other shoaling species, one or more individuals sometimes leave their school and approach a putative predator, in an attempt to gain information about this potential threat, a behavior which has been called "predator inspection" [158,159]; the information thus obtained seems to be actively or passively transmitted to noninspectors in the school [160]. Inspection behavior entails considerable risk taking; in guppies (*Poecilia reticulata*), increased tendency to inspect predators leads to increased predation [161]. In zebrafish, predator inspection is correlated with basal activity level in three different strains, but not with the frequency that individual fish left the shoal nor with the latency to feed after disturbance [138], suggesting that this trait is more related to activity than with anxiety.

Most behavioral work with zebrafish so far have used wild-type zebrafish, but one work analyzed different strains (TM1, SH and Nadia) in predator inspection [138]. Fish from the TM1 strain were consistently more likely to inspect predators, when tested in groups, than fish from the SH and Nadia zebrafish. These and other results regarding novel object inspection and bottom-dwelling (see below) parallel strain differences in anxiety-like behavior in rodents [162].

As with most other models, so far no anxiety-altering drugs were tested using predator avoidance or approach responses in zebrafish. However, given the possible confounds delineated above, and the lack of correlation with other measures of "emotionality" or "boldness", it is unlikely that antipredator behavior turns out to be useful as a model of anxiety in this species (but see [20], for a different view).

Another potential use is to expose zebrafish to predators (or partial predator stimuli) and observe posterior behavior in other anxiety tests. At least in rodents, increases in anxiety-like behavior after the presentation of predators or partial predator stimuli are a complex process. Adamec and Shallow [163] assessed rats in the hole board and elevated plus-maze after they were placed in a room with a cat or with the residual odor of a cat; rats exposed to this latter partial predator stimulus had increased frequencies and durations of risk assessment behavior in the elevated plus-

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maze, but animals which were exposed to cats showed reduced risk assessment. The same pattern is observed when testing animals in the presence of these threat stimuli: animals tested in the presence of cats show reduced risk assessment, while animals tested in the presence of cat odor show increased risk assessment—which reinforces the idea that risk assessment behavior is associated with "potential" threat.

Egan et al. [18] explored this possibility by exposing zebrafish for either 5 or 30-min to alarm substance, and latter assessed in the novel tank diving test (see below). Animals which were exposed to the alarm pheromone spent more time in the bottom of the tank and explored less its upper half, also presenting a higher frequency of freezing bouts and erratic movements. Interestingly, prolonged exposure (that is, exposure for 30 min) did not produce any significant change in these parameters, suggesting habituation [18,25,141].

3.2.2. Alarm reaction ("Schrecksreaktion")

The alarm reaction of Ostariophysan fishes to a substance released by the destruction of epidermal club cells in conspecifics was first described by von Frisch [164]. In its current form, alarm reaction is usually quantified as the frequency and duration of erratic movement and freezing episodes [22,131,140,165–167]. Coho salmons (*Oncorhynchus kisutch*, Salmonidae) exposed to alarm substance show elevations in plasma cortisol and glucose [168], as do zebrafish visually exposed to predators [14]. Alarm substances from other species of the Ostariophysian superorder were identified, and hypoxanthine 3-*N*-oxide (a purine derivative oxidized at the 3-position) was shown to be the common compound to all alarm substances [167,169,170]. This substance has been shown to dose-dependently increase the number of erratic movement episodes and the number of jumps in short-fin wild-type zebrafish, but did not increase the incidence of freezing or the proportion of the trial the animal spent in the bottom of the tank [140]. Interestingly, Suboski and co-workers [77,171] observed that the immediacy, depth and duration of change in the vertical distribution in the tank by fish are the best indexes of alarm.

The pharmacology of alarm reaction is rather inconsistent. In rodents, behavioral changes associated with exposure to predator odor are counteracted by benzodiazepines, but not serotonergic agents (with the exception of a weak effect of chronic administration of selective serotonin reuptake inhibitors)[156]. Fathead minnows (*Pimephales promelas*, Cyprinidae) exposed to alarm substance show increased cover use and decreased foraging, an effect which was reversed by pre-administration of the benzodiazepine receptor agonist chlordiazepoxide [172]; in the piauçu *Leporinus macrocephalus* (Anostomidae), chronic treatment with fluoxetine reduces alarm substance-induced bottom-dwelling (Augusto Barbosa Junior and Annette Hoffmann, personal communication). Thus, the potentiation of the anxiogenic properties of novelty by pre-exposure to alarm substance (or to predators, as for that matter) show potential as a model of stress-potentiated anxiety—akin to the effects of exposure to predator odors in rodents, which has been proposed as a model of posttraumatic stress disorder [164].

3.2.3. Open-field

The exploration of an open-field is a standard measure of emotionality, exploratory drive and habituation in rodents, albeit not without complications [173]. In fish, open-field (or "open-tank") exploration has also been used as a measure of anxiety-like behavior. Animals are transferred to a novel, usually round, tank, and its exploratory behaviors (including thigmotaxis, exploratory swimming, freezing bouts, and erratic movements) are recorded [25] (Table 4). In this task, zebrafish exhibit initially elevated levels of activity, which presents intra-session habituation [174]; aside from that aspect of non-associative learning, zebrafish also show aspects of fear [19,25,129,130,175–177], including "thigmotactic"/"centrophobic" behavior; as in rodents, then, zebrafish tends to spend the majority of time in a trial swimming near to the walls of the tank, making short-duration incursions into the center of the apparatus. Interestingly, fish with an atypical right-lateralized parapineal position spends less time in the periphery of an open-field than left-lateralized animals [178].

Pharmacological studies of open-field activity of zebrafish are more abundant than those using any of the other models. The blockade of histidine decarboxylase, the rate-limiting enzyme in the synthesis of histamine, by α-fluoromethylhistidine increases the time zebrafish spends swimming in the center of a circular open-field, suggesting an anxiolytic effect [177]. Likewise, cocaine withdrawal or the administration of the benzodiazepine receptor inverse agonist FG-7142 increases thigmotaxis, an effect which is counteracted by the administration of a non-sedative dose of diazepam [19]. Chronic fluoxetine exposure and acute ethanol both decrease the occurrence of erratic movements in the open-field [18,130]. In larval zebrafish, sub-chronic (24 h) exposure to the serotonin reuptake inhibitor fluoxetine between 4 and 5 days post-fertilization reduces total locomotion in the open-field [175]; these results, however, correlate with decreased expression of the 5-HT_{1A} receptor and the serotonin transporter in the spinal cord, but not in the brain, suggesting an effect on locomotor output alone. Similar results on locomotion in larval zebrafish (7 days postfertilization [dpf]) are observed with the administration of the dopaminergic antagonist clozapine, an effect which is countered by administration of the D4-selective agonist ABT-724 [179]. Ethanol, on the other hand, produces a marked hyperactive behavior in 6 dpf larvae [180].

As is the case with mammals and birds [173], pharmacological response to anxiolytic or anxiogenic drugs in the open-field is highly dependent on the measures which are recorded. The usual measure of thigmotaxis is probably more useful to analyze effects of compounds on anxiety than exploratory swimming or total distance travelled, as these latter measures are too prone to confounds with locomotor deficits. "Ethological" measures, such as the occurrence of freezing bouts or erratic swimming can also be targeted by these drugs, but results so far have not been consistent. Therefore, the open-field task remains an important option for the measurement of anxiety in zebrafish, but measures other than total distance travelled are necessary to control for locomotor effects.

3.2.4. Novel tank diving test

So far, the novel tank diving test is the most extensively studied model of anxiety in zebrafish, including pharmacological measures. This model exploits the observation that some fish (including zebrafish) tend to spend a 70–85% of the first minute in the bottom third of a novel tank, a preference that is reduced to approximately chance levels by the end of a 5- or 10-min test [18,22,130,133] (but see [129,132], for instances in which bottom-dwelling was not consistently observed). This behavior, called "bottom-dwelling", is sensitive to diazepam and buspirone, but not chlordiazepoxide [181]. It is also sensitive to chronic fluoxetine and acute ethanol treatments [18,25], but not to yohimbine (a heterogeneous pharmacon which has anxiogenic properties) treatment [182]. Acute citalopram (a selective serotonin reuptake inhibitor), ethanol (0.5%) or desipramine (which preferentially inhibits the uptake of noradrenaline and, to a lesser extent, inhibits the uptake of serotonin) treatment decreases bottom-dwelling in PETCO zebrafish [182]. Acute ethanol treatment, which affects $GABA_A$-R, NMDA-R, μ-opioid receptors, dopamine and norepinephrine receptors, and voltage-sensitive calcium receptors [183], shows an U-shaped dose-response profile in bottom-dwelling in the AB strain, but an inverted U-shaped profile in short-fin wild-type, long-fin wild-type, and leopard strains [151]; leopard zebrafish also seem to

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present increased baseline anxiety in this test, while WIK zebrafish spend most of the time in the top of the tank [18,25,182] (Table 4). Less "traditional" pharmacological manipulations include nicotine and nicotinic receptor antagonists, which show opposing effects on bottom-dwelling [133,176,182], and the NMDA-R antagonist MK-801, which decreases bottom-dwelling in zebrafish at its highest doses [16], an anxiolytic-like effect which is consistent with what has been observed in rodents [184–189]. Since MK-801 also increases locomotion and induces "circling" behavior (a putative stage of clonus-like convulsion in fish [13]) in larval and adult zebrafish [16,88,190], it is difficult to judge whether this represents a true anxiolytic effect or merely hyperlocomotion. Nicotine, on the other hand, seems to show an anxiogenic profile in AB, WIK and GloFish zebrafish, but an anxiolytic profile in PetCo ($25\,mg\,l^{-1}$) [182] and wild-type zebrafish ($50\,mg\,l^{-1}$) [133]; in wild-type zebrafish, the subtype selective antagonists methyllycaconitine (an α7 antagonist) and dihydro-β-erythroidine (an α4β2 antagonist) revert the anxiolytic effect of nicotine [176].

Exposure to an alarm substance has been shown to increase bottom-dwelling [18,25,77,171]; hypoxanthine 3-*N*-oxide, though, does not produce the same effect [140]. From a chemical standpoint, both the alarm substance and hypoxanthine 3-*N*-oxide have a purine skeleton as a structural component, and a nitrogen oxide as a functional component. In minnows (*P. promelas*) and daces (*Chrosomus neogaeus*), skin extract, hypoxanthine 3-*N*-oxide and pyridine-*N*-oxide increase shoaling, but structurally similar molecules without the nitrogen oxide functional group does not produce antipredator responses [191]. Interestingly, the xanthine caffeine, which contain the structural component but not the nitrogen oxide group, increases bottom-dwelling and decreases the number of transitions to the upper half of the novel tank [18], replicating part of the effects observed in rodents [192].

Some criticisms were raised regarding stimulus control in the novel tank diving test [17]. The main driving force in bottom-dwelling in this test seems to be novelty, as animals which were housed in tanks which were similar to the test apparatus (same size and dimensions) presented a different profile of bottom-dwelling and swimming speed than animals which were housed in tanks which were dissimilar to the test apparatus [181]. In a 5-min session, animals housed in the dissimilar (wide) tanks showed a gradual increase in bottom-dwelling and swimming speed, while animals housed in similar (narrow) tanks did not show this variation. However, it can be argued that the main difference between housing conditions is not necessarily its similitude to the test apparatus, but its dimensions (narrow versus wide), which could indicate an effect of environmental/social stress. Being housed in a narrower tank impairs habituation of novelty, an effect which parallels decreases in habituation rates of exploratory behavior in the open-field by rodents which were reared in social isolation [193,194]. In farmed Atlantic salmon, pre-spawning maternal cortisol levels dose-dependently increases bottom-dwelling in novel environments in offspring [195], an effect which has also been observed in rodents [196].

Zebrafish exposed to the novel tank for a 6- or a 30-min trial show increases in exploratory behavior and decreases in the frequency and duration of freezing, but no effects on erratic swimming [141]; when exposed to the apparatus for 7 consecutive days, animals show increases in exploratory behavior and decreases in freezing and erratic swimming [141]. These results are difficult to interpret, but suggest that novelty is the main controlling factor in this test. Intra- and inter-session habituation of locomotion is observed in the open-field test in rodents [197], but no habituation of open arms avoidance is observed in the elevated plus-maze [198] (consistent with the aversive character of this portion of the apparatus for rodents). In another test of anxiety, the scototaxis test, we observed intra- or inter-session habituation of locomotor activity, but not of the avoidance of the white compartment, using as inter-session interval either 15-min or 24-h. [139,142], which suggests that the white compartment is aversive to this species (see below). These differences between the novel tank diving test and the scototaxis test are important, as novelty seems to increase arousal, but not necessarily anxiety (which seems to be generated by motivational conflict), in animals [198–202].

One attempt which has been made to establish which behaviors are good indexes of anxiety was to expose fish to predators or partial predator stimuli, analyzing their ethogram after the exposure [15,18,22,129–132,137,203]. One of these behavioral measures is precisely bottom-dwelling. Exposure to three-dimensional predator models increase bottom-dwelling [129], but exposure to a computer animated model does not produce any changes in bottom-dwelling (but it also does not induce any avoidance responses) [132]. Also, increased bottom-dwelling after exposure to alarm substances or hypoxanthine 3-*N*-oxide is controversial [18,22,25,77,140,171]. As we argued above, though, exposure to predators or partial predator stimuli is more likely to generate fear, and not anxiety responses, and might be a poor predictor of anxiety in zebrafish. Still, the novel tank diving test lacks the analysis of stimulus control that increases construct validity of spontaneous exploration models of anxiety [51].

3.2.5. Scototaxis

An alternative to the novel tank diving and open-field tests is scototaxis, the marked preference for dark environments presented by many teleost fish, including zebrafish [21,23,24,182].

> Behavior in this task (i.e., activity in the white compartment) reflects a conflict between the preference of the animal for protected areas (e.g., black substrata, in a process of crypsis) and an innate motivation to explore novel environments. Crypsis in fish depends on the dorsal distribution of melanophores in the fish surface; the presence of dorsal and dorsally oriented melanophores tends to minimize refraction and reflection of light incident on the fish, reducing the effectiveness of visual information for predators (...). Appropriate cryptic coloration requires the appropriate background and behavior [preference for dark substrata, in the scototaxis test] [24].

Most animals initially enter the dark compartment, where they spend the initial two-thirds of the session before exploring the white compartment (Unpublished data). This initial preference seems to be determined by telencephalic structures, as telencephalic ablation abolishes initial preference and increases the time spent in the white compartment [204]. However, the preference for dark environments does not seem to be due to appetitive approach to the black compartment, as high-avoidant animals confined in the white compartment show increased freezing and erratic movement [17]. Animals confined for three consecutive 15-min periods to the white compartment ("flooding") before testing show increased burst swimming, thigmotaxis and freezing in the white compartment, but no change in avoidance of the white compartment or in locomotor activity [139]; the avoidance of the white compartment also does not seem to habituate after repeated exposures to the apparatus [139,142], which suggest that neither avoidance of the white compartment alone nor approach to the black compartment alone can explain the behavior of zebrafish in this task—confirming the hypothesis that, as in rodent models such as the elevated plus-maze and the light–dark transitions test [46,51], scototaxis in zebrafish reflects an approach-avoidance motivational conflict.

At the present moment, the main disadvantage of the scototaxis test in relation to the open-field or the novel tank diving test is that this test is not yet pharmacologically validated. Preliminary results from our laboratory and by Su Guo uncovered an anxiolytic effect of low doses of chlordiazepoxide [24,204], a compound which reduces

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theta frequency in the hippocampus of rats [205] and produces anxiolytic effects in the light–dark transitions box in mice [206–209] and in the cat odor challenge model in rats [210]; interestingly, chlordiazepoxide was not detected as an anxiolytic compound in the novel tank diving test [181]. Currently, we are conducting the pharmacological validation of the scototaxis test for zebrafish and goldfish, testing serotonergic drugs, benzodiazepines, ethanol, caffeine and amphetamines.

A variation of the scototaxis test, the aquatic light/dark plus-maze, was proposed by James Sackermans and colleagues [182]. This apparatus is composed of four opposing "arms", two with white substrata and two with black substrata, separated by a central compartment with grey substratum. Untreated fish tend to freeze in this central apparatus and thereafter enter the black arms; they present scototaxis, in the sense of spending more time in the black arms than in the white arms of the apparatus. The initial time spent in the central compartment (i.e., latency for first choice) varies with strain (Table 3), with AB zebrafish presenting greater latencies than Petco or GloFish [182]. Nicotine did not produce any significant effect on total locomotion or white avoidance in this modified scototaxis test, but acute ethanol and chlordiazepoxide increased the time spent in the white arms [182]. The acute exposure of zebrafish to acute citalopram (an selective serotonin reuptake inhibitor) or yohimbine (an α-adrenoceptor antagonist, and, to a lesser extent, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{2A}, 5-HT_{2B}, and D_2 receptor antagonist, and 5-HT_{1A} receptor partial agonist) do not produce an anxiogenic effect, though [182], which suggests that different processes are being modeled in the scototaxis and novel tank diving tests.

3.2.6. Inspection of a novel object

The inspection of a novel object is sometimes used as a test of "boldness" in zebrafish [203,211], showing differences between wild and laboratory populations and being diminished after exposure to an alarm substance [211]. The general procedure for this task involves an open-field and several objects; the set of objects can be singly or repeatedly presented, and the behavior of the animal is recorded [212]. In rodents, this task is commonly used to assess habituation and recognition memory [212], and *not* anxiety—if anything, rodents display a tendency to approach, and not avoid, novel objects.

In wild-caught and AB laboratory zebrafish, the total time spent within 1.5 body lengths of a novel object in a single 10-min presentation correlates with shoaling tendency, which suggests that at least some components of novel object inspection in zebrafish are related to anxiety [137,203]. In both this work and the work of Ogwang [211], the inspection of a novel object is understood as a proxy for predator inspection, probably because predator inspection and shoaling tendency are correlated in guppies [213]. However, no correlation between predator inspection and shoaling tendency was observed in three different strains of zebrafish [138]. Also, we are not aware of any work in which the correlation between predator inspection and novel object inspection is tested. All these problems appreciably reduce the construct validity of novel object inspection as a model of anxiety in zebrafish. Another problem regards pharmacological validation, which is null in this model.

3.2.7. Shoaling

Zebrafish are a naturally shoaling species [136,137,214,215], a tendency which, in fish, is associated with decreased predation risk [216–218]. As in other responses associated with anxiety, shoaling is thought to be under a double constraint, as shoal density is inversely related to foraging efficacy [219,220] but directly related to predator avoidance [216,217]. There are considerable strain differences in shoaling tendency (Table 4); for example, SH zebrafish are much more likely to leave a shoal than Nadia and TM1 animals. In SH and TM1 zebrafish, leaving shoal is correlated with general activity level, while there is no correlation between both measures in Nadia zebrafish [138]; on the other hand, the less active Nadia zebrafish show a positive correlation between leaving shoal and predator inspection, a relationship that is not observed in both TM1 and SH animals [138]. In both AB laboratory zebrafish and wild animals captured in Bangladesh, shoaling tendency correlates with the total time spent near a novel object [203]; wild populations show no difference in their shoaling tendency, though [137]. Zebrafish increase the cohesion of their shoals in response to alarm substance [22,135] and to aerial predator models [136].

Many different methods have been proposed to measure shoaling in fish [129,136,137,221]; for example, it is possible to measure shoaling tendency by placing fish in the central compartment of a test tank flanked by adjacent compartments which either contain a conspecific shoal or is empty [137,203,222]. Another possibility is to measure the "cohesion" of a shoal by measuring, for each fish, its distance from its nearest neighbor [136,223]. With the exception of shoaling responses to predator models and alarm substance [22,135,136], few other potentially anxiogenic stimuli have been tested as elicitors of shoaling behavior. Novelty does not seem to control shoaling, as nearest neighbor distance does not decrease within or between sessions [136]; however, Wright and colleagues reported that there was a significant decrease in the time zebrafish from wild-derived populations (Nepal, Tangail, Santal and Canal) spent shoaling, with no differences between populations [137]. These conflicting results are probably due to differences in methodology.

So far, few pharmacological studies assessed shoaling. Ethanol reduced shoaling at high doses (0.5 and 1.0% v/v) in both acute exposition and chronic, developmental exposition in adult zebrafish [130,224]. The NMDA antagonist MK-801 dose-dependently disrupts shoaling in wild-type zebrafish [16]. In the medaka *Oryzias latipes*, diazepam decreased shoaling tendency [225]. These results are difficult to interpret, since MK-801 usually presents an anxiolytic-like profile [184,186,187]—and thus should decrease shoaling. It should be noted that, while shoaling behavior certainly involves defense reactions, other variables are probably involved; therefore, one could argue that the glutamatergic system has no role in controlling shoaling. Further work (perhaps with serotonergic agents) is necessary to better understand the potential of this model as an anxiety test.

3.3. Analysis of neurovegetative responses in zebrafish

Even in rodents (Table 1), reports of measures of behavioral, neuroendocrine and autonomic arousal are uncommon in models of anxiety. The more general term "arousal" refers to behavior that is caused by exposure to an unfamiliar stimulus, such as a novel environment or sudden noise [49]. For example, Morato and colleagues have used three types of elevated plus-maze (normal, with no walls, and with walls in all arms), reporting that the 'deactivated' plus-maze (with walls in all arms) generates behavior that is related to arousal but not anxiety [201]; the 'deactivated' plus-maze does not induce serotonin release in the ventral hippocampus of rats as does the normal plus-maze [226]. In the scototaxis test, an all-black tank has been used in some articles to control for novelty-induced arousal [21,23,24], and both high-avoidance and low-avoidance animals swim more and display more thigmotaxis while confined in the black compartment than in the white compartment of the scototaxis tank [17].

Behavioral responses associated with increased arousal are not easily dissociable in any species [49]. In fish, ventilatory frequency (opercular beats) has been proposed as a measure of novelty- or stress-induced arousal [227]. Briefly, when introduced in novel environments, fish show an increase in the frequency of opercular

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beats which is observed more radically after confinement stress or after exposure to predators [227–229]. In goldfish, opercular beat frequency has been reported to increase after presentation of a light stimulus at 2-min interval, which points to arousal [230]. On the other hand, in Nile tilapia (*Oreochromis niloticus*), opercular beat frequency increases by the same magnitude after handling and confinement stress, which reduces its reliability as a measure of stress versus arousal [228].

Another measure which has been used successfully in zebrafish responses to stress is whole-body cortisol levels [231], which are increased after direct or visual contact with a predator [14] and morphine or ethanol withdrawal [144], and decreased after chronic fluoxetine treatment [18,25]. There is little evidence, however, that cortisol responses are associated with arousal in animal models or in human clinical settings [127], and they are more likely to represent an acute neuroendocrine response to stressful events (including fear-eliciting events) than arousal. Nonetheless, cortisol assays remain useful measures of the physiological adjustments associated with aversive events.

4. Conclusions

Almost 10 years ago, Robert Gerlai noted the lack of behavioral studies using zebrafish, a species which showed great potential as a behavioral genetic model [174]. In the last decade, many different models of anxiety and fear in zebrafish appeared, and it is now becoming clear that this species shows great potential as a pharmacological tool [232]. The ease at which mutations can be induced in zebrafish, combined with high-throughput screening assays, can quickly put this teleost in the limelight of behavioral neuroscience and neuropsychopharmacology.

We should be careful, though, to avoid making the same mistakes which were made on behavioral models of anxiety disorders in rodents. For example, it was not until the introduction of "ethopharmacological" analysis that buspirone, the 5-HT_{1A} agonist which was previously "missed" in behavioral screening of anxiolytic compounds, was detected as having anti-anxiety effects in rodents [38,60]. This example calls attention to the need for stricter and more careful "behavioral validation" of models of anxiety in order to define which endpoints are more sensitive to which pharmacological treatments, and especially to define if a given test *actually models what it is supposed to model* [51]. This is the sense with which McNaughton and Zangrossi Jr. stated that, at its limit, both predictive and face validity reduce to construct validity [51], *especially* in the case of anxiety models [52]. If we examine, for example, results regarding the pharmacology of the novel tank diving test [18,25,133,141,144,176,181], inconsistencies on the effects of pharmaceuticals on different behavioral endpoints on this test arise, and the considerable amount of "borderline" significant effects raises the possibility that these different endpoints are mediated by different motivational systems (e.g., fear versus anxiety). Careful parametric manipulation, such as what is being done in the scototaxis test [17,24,142,143], is slowly being applied to the novel tank diving test as well [141,181] and might shed some light on this question. Other models, such as open-field and antipredatory responses, would surely benefit from a similar approach. We insist, however, that model building and refining is a constant process, and the definition of appropriate endpoints and procedural parameters are never enough to state that a given model is "finished"; its further refinement is an ongoing activity.

It should be noted that building and refining should not be restricted to determining what endpoints are more sensitive to drug treatments. A great deal of research in the field of modeling regards the improvement of construct validity, and for that determining endpoints predictive of drug action is not enough. For example, in the scototaxis test it has been determined that stimulus control in the task is probably due to approach-avoidance conflict, and not solely by approach to the dark compartment or avoidance of the white compartment [139]; also, since locomotion but not white avoidance decreases after repeated exposure, novelty does not seem to be the main controlling variable [139,142]. These results suggest that behavior in the scototaxis test represents anxiety, and not fear or arousal. On the other hand, even though pharmacological research in the novel tank diving test are much more extensive than in any other model and the behavioral endpoints which are more sensitive to anxiolytic or anxiogenic manipulations are well defined, little is known about what are the controlling stimuli in this task. Therefore, there is still much work to be done in the field regarding parametric analyses to define just *what* the models are modeling.

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References

[1] Grunwald DJ, Eisen JS. Headwaters of the zebrafish—emergence of a new model vertebrate. Nat Rev Genet 2002;3:717–24.
[2] Barros TB, Alderton WK, Reynolds HM, Roach AG, Berghmans S. Zebrafish: an emerging technology for *in vivo* pharmacological assessment to identify potential safety liabilities in early drug discovery. Br J Pharmacol 2008;154:1400–13.
[3] Best JD, Alderton WK. Zebrafish: an in vivo model for the study of neurological diseases. Neuropsych Disease Treat 2008;4:567–76.
[4] Goldsmith P, Solari R. The role of zebrafish in drug discovery. Drug Discovery World 2003;(Spring):74–8.
[5] Nicolson T. Using zebrafish to study neuroscience. In: Nicolson T, editor. SfN short courses. Atlanta: Society for Neuroscience; 2006. p. 60.
[6] Ninkovic J, Bally-Cuif L. The zebrafish as a model system for assessing the reinforcing properties of drugs of abuse. Methods 2006;39:262–74.
[7] Panula P, Sallinen V, Sundvik M, Kolehmainen J, Torkko V, Tiitula A, et al. Modulatory neurotransmitter systems and behavior: towards zebrafish models of neurodegenerative diseases. Zebrafish 2006;3:235–47.
[8] Linney E, Upchurch L, Donerly S. Zebrafish as a neurotoxicological model. Neurotoxicol Teratol 2004;26:709–18.
[9] Jekosch K. The zebrafish genome project: sequence analysis and annotation. In: Detrich HW, Westerfield M, Zon LI, editors. The zebrafish: genetics, genomics and informatics. St. Louis: Academic Press; 2004. p. 686.
[10] Mathavan S, Lee SGP, Mak A, Miller LD, Murthy KRK, Govindarajan KR, et al. Transcriptome analysis of zebrafish embryogenesis using microarrays. PLoS Genet 2005;1:e29.
[11] Lucitt MB, Price TS, Pizarro A, Wu W, Yocum AK, Seiler C, et al. Analysis of the zebrafish proteome during embryonic development. Mol Cell Proteomics 2008;7:981–94.
[12] Xu X, Scott-Scheiern T, Kempker L, Simons K. Active avoidance conditioning in zebrafish (*Danio rerio*). Neurobiol Learn Mem 2007;87:72–7.
[13] Baraban SC, Taylor MR, Castro PA, Baier H. Pentylenetetrazole induced changes in zebrafish behavior, neural activity and c-fos expression. Neuroscience 2005;131:759–68.
[14] Barcellos LJG, Ritter F, Kreutz LC, Quevedo RM, Silva LB, Bedin AC, et al. Whole-body cortisol increases after direct and visual contact with a predator in zebrafish, *Danio rerio*. Aquaculture 2007;272:774–8.
[15] Bass SL, Gerlai R. Zebrafish (*Danio rerio*) responds differentially to stimulus fish: the effects of sympatric and allopatric predators and harmless fish. Behav Brain Res 2008;186:107–17.
[16] Echevarria DJ, Hammack CM, Pratt DW, Hosemann JD. A novel test battery to assess global drug effects using the zebrafish. Int J Comp Psychol 2008;21:19–34.
[17] Blaser RE, Chadwick L, McGinnis GC. Behavioral measures of anxiety in zebrafish (*Danio rerio*). Behav Brain Res; in press.
[18] Egan RJ, Bergner CL, Hart PC, Cachat JM, Canavello PR, Elegante MF, et al. Understanding behavioral and physiological phenotypes of stress and anxiety in zebrafish. Behav Brain Res 2009;205:38–44.
[19] López-Patiño MA, Yu L, Cabral H, Zhdanova IV. Anxiogenic effects of cocaine withdrawal in zebrafish. Physiol Behav 2008;93:160–71.
[20] Gerlai R. Zebrafish antipredatory responses: a future for translational research? Behav Brain Res 207:223–31:in press.
[21] Serra EL, Medalha CC, Mattioli R. Natural preference of zebrafish (*Danio rerio*) for a dark environment. Braz J Med Biol Res 1999;32:1551–3.

[22] Speedie N, Gerlai R. Alarm substance induced behavioral responses in zebrafish (*Danio rerio*). Behav Brain Res 2008;188:168–77.
[23] Maximino C, Brito TM, Moraes FD, Oliveira FVC, Taccolini IB, Pereira PM, et al. A comparative analysis of the preference for dark environments in five teleosts. Int J Comp Psychol 2007;20:351–67.
[24] Maximino C, Brito TM, Dias CAGM, Gouveia A, Jr., Morato S. Scototaxis as anxiety in fish. Nat Protoc 2009;4:in press.
[25] Cachat JM, Canavello PR, Elegante MF, Bartels BK, Elkhayat SI, Hart PC, et al. Modeling stress and anxiety in zebrafish. In: Kalueff AV, Cachat J, editors. Zebrafish models in neurobehavioral research. New York: Humana Press; in press.
[26] Willner P. Introduction. In: Willner P, editor. Behavioral models in psychopathology: theoretical, industrial and clinical perspectives. Cambridge: Cambridge University Press; 1991.
[27] Bakshi VP, Kalin NH. Animal models and endophenotypes of anxiety and stress disorders. In: Davis KL, Charney D, Coyle JT, Nemeroff C, editors. Neuropsychopharmacology: the fifth generation of progress. American College of Neuropsychopharmacology; 2002.
[28] Gould TD, Gottesman II. Psychiatric endophenotypes and the development of valid animal models. Genes Brain Behav 2006;5:113–9.
[29] Kaplan A. The conduct of inquiry. Methodology for behavioral science. Aylesbury: International Textbook Company, Ltd.; 1973.
[30] Kalueff AV, Tuohimaa P. Experimental modeling of anxiety and depression. Acta Neurobiol Exp (Warsaw) 2004;64:439–48.
[31] LaFollette H, Shanks N. Two models of models in biomedical research. Phil Quart 1995;45:141–60.
[32] McKinney WT. Animal models of depression: an overview. Psych Dev 1984;2:77–96.
[33] van der Staay FJ. Animal models of behavioral dysfunctions: basic concepts and classifications, and an evaluation strategy. Brain Res Rev 2006;52:131–59.
[34] Willner P. Validation criteria for animal models of human mental disorders: learned helplessness as a paradigm case. Progr Neuro-Psychopharmacol Biol Psych 1986;10:677–90.
[35] McKinney WT, Bunney WE. Animal model of depression: I. Review of evidence: implications for research. Arch Gen Psych 1969;21:240–8.
[36] Holmes PV. Rodent models of depression: reexamining validity without anthropomorphic interference. Crit Rev Neurobiol 2003;15:143–74.
[37] Kalin NH, Shelton SE, Davidson RJ. Cerebrospinal fluid corticotropin-releasing hormone levels are elevated in monkeys with patterns of brain activity associated with fearful temperament. Biol Psych 2000;47:579–85.
[38] Rodgers RJ, Cao B-J, Holmes A. Animal models of anxiety: an ethological perspective. Braz J Med Biol Res 1997;30:289–304.
[39] Blanchard RJ, Yudko EB, Rodgers RJ, Blanchard DC. Defense system psychopharmacology: an ethological approach to the pharmacology of fear and anxiety. Behav Brain Res 1993;58:155–65.
[40] Blanchard DC, Blanchard RJ. Ethoexperimental approaches to the biology of emotion. Ann Rev Psychol 1988;39:43–68.
[41] Godsil BP, Tinsley MR, Fanselow MS. Motivation. In: Healy AF, Proctor RW, editors. Handbook of psychology. Hoboken: John Wiley & Sons; 2003. p. 33–60.
[42] Fanselow MS. Neural organization of the defensive behavior system responsible for fear. Psychonomic Bull Rev 1994;1:429–38.
[43] Fanselow MS, Lester L. A functional behavioristic approach to aversively motivated behavior: predatory imminence as a determinant of the topography of defensive behavior. In: Bolles RC, Beecher MD, editors. Evolution and learning. Hillsdale: Erlbaum; 1988. p. 185–211.
[44] Misslin R. The defense system of fear: behavior and neurocircuitry. Neurophysiol Clin 2003;33:55–66.
[45] Kavaliers M, Choleris E. Antipredator responses and defensive behavior: ecological and ethological approaches for the neurosciences. Neurosci Biobehav Rev 2001;25:577–86.
[46] McNaughton N, Corr PJ. A two-dimensional neuropsychology of defense: fear/anxiety and defensive distance. Neurosci Biobehav Rev 2004;28:285–305.
[47] Catherall DR. How fear differs from anxiety. Traumatology 2003;9:76–92.
[48] Blanchard DC, Blanchard RJ. Defensive behaviors, fear, and anxiety. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B.V; 2008. p. 63–79.
[49] Toth M, Zupan B. Neurobiology of anxiety. In: Sibley DR, Hanin I, Kuhar M, Skolnick P, editors. Handbook of contemporary neuropharmacology. Nova Iorque: John Wiley & Sons; 2007. p. 3–58.
[50] Levine S, Madden J, Conner RL. Physiological and behavioral effects of prior aversive stimulation (preshock) in the rat. Physiol Behav 1973;10:467–71.
[51] McNaughton N, Zangrossi Jr H. Theoretical approaches to the modeling of anxiety in animals. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B.V; 2008. p. 11–27.
[52] Maximino C, Brito TM, Gouveia A, Jr. Construct validity of behavioral models of anxiety: where experimental psychopathology meets ecology and evolution. Psychol Neurosci; in press.
[53] Blanchard RJ, Blanchard DC. An ethoexperimental analysis of defense, fear and anxiety. In: McNaughton N, Andrews G, editors. Anxiety. Dunedin: Otago University Press; 1990. p. 124–33.
[54] Davidson MK, Lindsey JR, Davis JK. Requirements and selection of an animal model. Israel J Med Sci 1987;23:551–5.
[55] Geyer MA, Markou A. The role of preclinical models in the development of psychotropic drugs. In: Davis KL, Charney D, Coyle JT, Nemeroff C, editors. Neuropsychopharmacology: the fifth generation of progress. New York: Raven Press/American College of Neuropsychopharmacology; 2002. p. 445–55.
[56] Matthysse S. Animal models in psychiatric research. Prog Brain Res 1986;65:259–70.
[57] Ratner SC. Comparative aspects of hypnosis. In: Gordon JE, editor. Handbook of clinical and experimental hypnosis. New York: Macmillan; 1967. p. 550–87.
[58] Graeff FG. Neuroanatomy and neurotransmitter regulation of defensive behaviors and related emotions in mammals. Braz J Med Biol Resea 1994;27:811–29.
[59] Blanchard RJ, Griebel R, Henrie JA, Blanchard DC. Differentiation of anxiolytic and panicolytic drugs by effects on rat and mouse defense test batteries. Neurosci Biobehav Rev 1997;21:783–9.
[60] Rodgers RJ. Animal models of 'anxiety': where next? Behav Pharmacol 1997;8:477–96 [discussion 497–504].
[61] Fanselow MS, Possusamy R. The use of conditioning tasks to model fear and anxiety. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B. V; 2008. p. 29–47.
[62] Litvin Y, Pentkowski NS, Pobbe RL, Blanchard DC, Blanchard RJ. Unconditioned models of fear and anxiety. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B. V; 2008. p. 81–99.
[63] Panksepp J. Affective neuroscience: the foundations of human and animal emotions. Nova Iorque: Oxford University Press; 1998.
[64] Canteras NS, Blanchard DC. A behavioral and neural systems comparison of unconditioned and conditioned defensive behavior. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B.V.; 2008. p. 141–53.
[65] Pavlov IP. Conditioned reflexes. London: Oxford University Press; 1927.
[66] Ferster CB, Skinner BF. Schedules of reinforcement. New York: Appleton-Century-Crofts; 1957.
[67] Dinsmoor JA. Punishment. In: O'Donohue WT, editor. Learning and behavior therapy. Boston: Allyn & Bacon; 1998. p. 188–204.
[68] Hineline PN. Negative reinforcement and avoidance. In: Honig WK, Staddon JER, editors. Handbook of operant behavior. Englewood Cliffs: Prentice Hall; 1977. p. 364–414.
[69] Dinsmoor JA, Sears GW. Control of avoidance by a response-produced stimulus. Learn Motiv 1973;4:284–93.
[70] Dinsmoor JA. Escape, avoidance, punishment: where do we stand? J Exp Anal Behav 1977;28:83–95.
[71] Dinsmoor JA. Stimuli inevitably generated by behavior that avoids electric shock are inherently reinforcing. J Exp Anal Behav 2001;75:311–33.
[72] Overmier JB, Hollis KL. Fish in the think tank: learning, memory and integrated behavior. In: Kesner RP, Olton DS, editors. Neurobiology of comparative cognition. Hillsdale: Lawrence Erlbaum Associates; 1990. p. 204–36.
[73] Davey G. Comparative aspects of conditioning: Pavlovian learning. In: Davey G, editor. Theory El. London: Routledge; 1989. p. 23–57.
[74] Rodríguez F, Duran E, Gomez A, Ocana FM, Avarez E, Jiménez-Moya F, et al. Cognitive and emotional functions of the teleost fish cerebellum. Brain Res Bull 2005;66:365–70.
[75] Álvarez E, Gómez A, Durán E, Ocaña FM, Jiménez-Moya F, Broglio C. Brain substrates of "eye-blink" classical conditioning in goldfish. Acta Neurobiol Exp [Warsaw] 2003;63:62.
[76] Yoshida M, Okamura I, Uematsu K. Involvement of the cerebellum in classical fear conditioning in goldfish. Behav Brain Res 2004;153:143–8.
[77] Hall D, Suboski MD. Visual and olfactory stimuli in learned release of alarm reactions by zebra danio fish (*Brachydanio rerio*). Neurobiol Learn Mem 1995;63:229–40.
[78] Blanchard RJ, Yang M, Li CI, Gervacio A, Blanchard DC. Cue and context conditioning of defensive behaviors to cat odor stimuli. Neurosci Biobehav Rev 2001;25:587–95.
[79] Dielenberg RA, McGregor IS. Habituation of the hiding response to cat odor in rats (*Rattus norvegicus*). J Comp Psychol 1999;113:376–87.
[80] McGregor IS, Dielenberg RA. Differential anxiolytic efficacy of a benzodiazepine on first versus second exposure to a predatory odor in rats. Psychopharmacology 1999;147:174–81.
[81] File SE. The interplay of learning and anxiety in the elevated plus-maze. Behav Brain Res 1993;58:199–202.
[82] Davis M. Pharmacological and anatomical analysis of fear conditioning. NIDA Res Monogr 1990;97:126–62.
[83] Mowrer OH. On the dual nature of learning: a reinterpretation of "conditioning" and "problem-solving". Harvard Educ Rev 1947;17:102–48.
[84] Schoenfeld WN. An experimental approach to anxiety, escape and avoidance behavior. In: Hoch PJ, Zubin J, editors. Anxiety. New York: Grune & Stratton; 1950. p. 70–99.
[85] Sidman M. Avoidance conditioning with brief shock and no exteroceptive warning signal. Science 1953;46:253–61.
[86] Sidman M. Some properties of the warning stimulus in avoidance behavior. J Comp Physiol Psychol 1955;48:444–50.
[87] Blank M, Guerim LD, Cordeiro RF, Vianna MRM. A one-trial inhibitory avoidance task to zebrafish: rapid acquisition of an NMDA-dependent long-term memory. Neurobiol Learn Mem 2009;92:529–34.
[88] Swain HA, Sigstad C, Scalzo FM. Effects of dizocilpine (MK-801) on circling behavior, swimming activity, and place preference in zebrafish (*Danio rerio*). Neurotoxicol Teratol 2004;26:725–9.
[89] Nam RH, Kim W, Lee CJ. NMDA receptor-dependent long-term potentiation in the telencephalon of the zebrafish. Neurosci Lett 2004;370:248–51.

UNCORRECTED PROOF

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[90] Quevedo J, Vianna M, Zanatta MS, Roesler R, Izquierdo I, Jerusalinsky D, et al. Involvement of mechanisms dependent on NMDA receptors, nitric oxide and protein kinase A in the hippocampus but not in the caudate nucleus in memory. Behav Pharmacol 1997;8:713–7.
[91] Xu X, Davis RE. *N*-Methyl-D-aspartate receptor antagonist MK-801 impairs learning but not memory fixation or expression of classical fear conditioning in goldfish (*Carassius auratus*). Behav Neurosci 1992;106:307–14.
[92] Xu X, William B, Bridget L, Spranger J. Comparison of the amnestic effects on NMDA receptor antagonist MK-801 and nitric oxide synthase inhibitors: L-NAME and L-NOARG in goldfish. Behav Neurosci 1998;112:892–9.
[93] Zhdanova IV, Wang S, Yu L, Quasarano C. Sleep deprivation affects learning and/or memory in adult zebrafish. In: 32nd annual meeting of the society for neuroscience. 2002. p. 371.314.
[94] Kloss JD, Szuba MP, Dinges DF. Sleep loss and sleepiness: physiological and neurobehavioral effects. In: Davis KL, Charney D, Coyle JT, Nemeroff C, editors. Neuropsychopharmacology: the fifth generation of progress. New York: Raven Press/American College of Neuropsychopharmacology; 2002. p. 1895–905.
[95] Flood NC, Overmier JB, Savage GE. Teleost telencephalon and learning: an interpretive review of data and hypotheses. Physiol Behav 1976;16:783–98.
[96] Portavella M, Torres B, Salas C. Avoidance response in goldfish: emotional and temporal involvement of medial and lateral telencephalic pallium. J Neurosci 2004;24:2335–42.
[97] Salas C, Broglio C, Durán E, Gómez A, Ocaña FM, Jiménez-Moya F, et al. Neuropsychology of learning and memory in teleost fish. Zebrafish 2006;3:157–71.
[98] Rupp B, Reichert H, Wullimann MF. The zebrafish brain: a neuroanatomical comparison with the goldfish. Anat Embryol 1996;194:187–203.
[99] Bovet D, Bovet-Nitti F, Oliverio A. Effects of nicotine on avoidance conditioning of inbred strains of mice. Psychopharmacology 1966;10:1–5.
[100] Bironi JD, Arneric SP. Behav Neural Biol 1993;59.
[101] Martí Baros D, Ramirez MR, Reis EA, Izquierdo I. Participation of hippocampal nicotinic receptors in acquisition, consolidation and retrieval of memory for one trial inhibitory avoidance in rats. Neuroscience 2004;126:651–6.
[102] Levin ED, Limpuangthip J, Rachakonda T, Peterson M. Timing of nicotine effects on learning in zebrafish. Psychopharmacology 2006;184:547–52.
[103] Levin ED, Chen E. Nicotinic involvement in memory function in zebrafish. Neurotoxicol Teratol 2004;26:731–5.
[104] Eddins D, Petro A, Williams P, Cerutti DT, Levin ED. Nicotine effects on learning in zebrafish: the role of dopaminergic systems. Psychopharmacology 2009;202:103–9.
[105] Faganello FR, Medalha CC, Mattioli R. Haloperidol and chlorpheniramine interaction in inhibitory avoidance in goldfish. Behav Brain Res 2003;147:83–8.
[106] Rossato JI, Bevilaqua LRM, Izquierdo I, Medina JH, Cammarota M. Dopamine controls persistence of long-term memory storage. Science 2009;325:1017–20.
[107] Pradel G, Schnachner M, Schmidt R. Inhibition of memory consolidation by antibodies against cell adhesion molecules after active avoidance conditioning in zebrafish. J Neurobiol 1999;39:197–206.
[108] Pradel G, Schmidt R, Schnachner M. Involvement of L1.1 in memory consolidation after active avoidance conditioning in zebrafish. J Neurobiol 2000;43:389–403.
[109] Stork O, Welzl H, Wotjak CT, Hoyer D, Delling M, Cremer H, et al. Anxiety and increased 5-HT1A receptor response in NCAM null mutant mice. J Neurobiol 1999;40:343–55.
[110] Law JW, Lee AY, Sun M, Nikonenko AG, Chung SK, Dityatev A, et al. Decreased anxiety, altered place learning, and increased CA1 basal excitatory synaptic transmission in mice with conditional ablation of the neural cell adhesion molecule L1. J Neurosci 2003;23:10419–32.
[111] Rose SP. Cell adhesion molecules and the transition from short- to long-term memory. J Physiol 1996;90:387–91.
[112] Mineka S. The role of fear in theories of avoidance learning, flooding, and extinction. Psychol Bull 1979;86:985–1010.
[113] Fernández-Teruel A, Escorihuela RM, Núñez JF, Zapata A, Boix F, Salazar W, et al. The early acquisition of two-way (shuttle-box) avoidance as an anxiety-mediated behavior: psychopharmacological validation. Brain Res Bull 1991;26:173–6.
[114] Mauthner L. Untersuchungen über den Bau des Rückenmarkes der Fische. Sitz ber Kgl Preuss Wiss 1859;34:31–6.
[115] Dill LM. The escape response of the Zebra Danio (*Brachydanio rerio*). I. The stimulus for escape. Anim Behav 1974;22:711–22.
[116] Eaton RC, Nissanov J. A review of Mauthner-initiated escape behaviour and its possible role in hatching in the immature zebrafish, *Brachydanio rerio*. Environ Biol Fishes 1985;12:265–79.
[117] Eaton RC, Bombardieri RA, Meyer DL. The Mauthner-initiated startle response in teleost fish. J Exp Biol 1977;66:65–81.
[118] Korn H, Faber DS. The Mauthner cell half a century later: a neurobiological model for decision-making? Neuron 2005;47:13–28.
[119] Eaton RC, Lee RKK, Foreman MB. The Mauthner cell and other identified neurons of the brainstem escape network of fish. Prog Neurobiol 2001;63:467–85.
[120] Marino-Neto J, Sabbatini RME. Discrete telencephalic lesions accelerate the habituation rate of behavioral arousal responses in Siamese fighting fish (*Betta splendens*). Braz J Med Biol Res 1983;16:271–8.
[121] Collins LE, Waldeck RF. Telencephalic ablation results in decreased startle response in goldfish. Brain Res 2006;1111:162–5.
[122] Best JD, Berghmans S, Hunt JJFG, Clarke SC, Fleming A, Goldsmith P, et al. Non-associative learning in larval zebrafish. Neuropsychopharmacology 2008;33:1206–15.
[123] Granato M, van Eeden FJ, Schach U, Trowe T, Brand M, Furutani-Seiki M, et al. Genes controlling and mediating locomotion behavior of the zebrafish embryo and larva. Development 1996;123:399–413.
[124] Lorent K, Liu KS, Fetcho JR, Granato M. The zebrafish space cadet gene controls axonal pathfinding of neurons that modulate fast turning movements. Development 2001;128:2131–42.
[125] van der Kolk BA. Psychobiology of posttraumatic stress disorder. In: Panksepp J, editor. Textbook of biological psychiatry. Hoboken: Wiley-Liss; 2004. p. 319–44.
[126] Dlugos CA, Rabin RA. Ethanol effects on three strains of zebrafish: model system for genetic investigations. Pharmacol Biochem Behav 2003;74:471–80.
[127] Lang PJ, Davis M, Öhman A. Fear and anxiety: animal models and human cognitive psychophysiology. J Affect Disorders 2000;61:137–59.
[128] Cruz APM, Frei F, Graeff FG. Ethopharmacological analysis of rat behavior on the elevated plus-maze. Pharmacol Biochem Behav 1994;49.
[129] Blaser RE, Gerlai R. Behavioral phenotyping in zebrafish: comparison of three behavioral quantification methods. Behav Res Methods 2006;38:456–69.
[130] Gerlai R, Lahav M, Guo S, Rosenthal A. Drinks like a fish: Zebra fish (*Danio rerio*) as a behavior genetic model to study alcohol effects. Pharmacol Biochem Behav 2000;67:773–82.
[131] Gerlai R, Lee V, Blaser RE. Effects of acute and chronic ethanol exposure on the behavior of adult zebrafish (*Danio rerio*). Pharmacol Biochem Behav 2006;85:752–61.
[132] Gerlai R, Fernandes Y, Pereira T. Zebrafish (*Danio rerio*) responds to the animated image of a predator: towards the development of an automated aversive task. Behav Brain Res 2009;201:318–24.
[133] Levin ED, Bencan Z, Cerutti DT. Anxiolytic effects of nicotine in zebrafish. Physiol Behav 2007;90:54–8.
[134] Lockwood B, Bjerke S, Kobayashi K, Guo S. Acute effects of alcohol on larval zebrafish: a genetic system for large-scale screening. Pharmacol Biochem Behav 2004;77:647–54.
[135] Rehnberg BG, Smith RJF. The influence of alarm substance and shoal size on the behaviour of zebra danios (*Brachydanio rerio*) (Cyprinidae). J Fish Biol 1988;33:155–63.
[136] Miller N, Gerlai R. Quantification of shoaling behaviour in zebrafish (*Danio rerio*). Behav Brain Res 2007;184:157–66.
[137] Wright D, Rimmer LB, Pritchard VL, Krause J, Butlin RK. Inter and intra-population variation in shoaling and boldness in the zebrafish (*Danio rerio*). Naturwissenschaften 2003;90:374–7.
[138] Moretz JA, Martins EP, Robison BD. Behavioral syndromes and the evolution of correlated behavior in zebrafish. Behav Ecol 2007;18:556–62.
[139] Maximino C, Brito TM, Colmanetti R, Pontes AAA, Castro HM, Lacerda RIT, et al. Parametric analyses of anxiety in zebrafish scototaxis. Behav Brain Res 2010;210:1–7.
[140] Parra KV, Adrian Jr JC, Gerlai R. The synthetic substance hypoxanthine 3-N-oxide elicits alarm reactions in zebrafish (*Danio rerio*). Behav Brain Res 2009;205:336–41.
[141] Wong K, Elegante M, Bartels B, Elkhayat S, Tien D, Roy S, et al. Analyzing habituation responses to novelty in zebrafish (*Danio rerio*). Behav Brain Res; in press.
[142] Maximino C, Brito TM, Colmanetti R, Castro HM, Lacerda RIT, Pontes AAA, et al. Intersession habituation of locomotion and white avoidance in the scototaxis test in zebrafish, goldfish, and yellow-tailed lambari, using a 24-h intersession interval. Anim Behav; submitted for publication.
[143] Maximino C, Brito TM, Colmanetti R, Pontes AAA, Castro HM, Lacerda RIT, et al. Anxiogenic stimuli in zebrafish dark/light preference test. Behav Brain Res; submitted for publication.
[144] Cachat J, Cavanello P, Elegante M, Bartels B, Hart P, Bergner C, et al. Modeling withdrawal syndrome in zebrafish. Behav Brain Res; in press.
[145] Brandão ML, Zanoveli JM, Ruiz-Martinez RC, Oliveira LC, Landeira-Fernandez J. Different patterns of freezing behavior organized in the periaqueductal gray of rats: association with different types of anxiety. Behav Brain Res 2008;188:1–13.
[146] Mikheev VN, Andreev OA. Two-phase exploration of a novel environment in the guppy, *Poecilia reticulata*. J Fish Biol 1993;42:375–83.
[147] Guo S. Using zebrafish to assess the impact of drugs on neural development and function. Expert Op Drug Discov 2009;4:712–26.
[148] López-Patiño MA, Yu L, Yamamoto BK, Zhdanova IV. Gender differences in zebrafish responses to cocaine withdrawal. Physiol Behav 2008;95:36–47.
[149] Bjerke S. Developing behavioral assays to study dopamine-related disorders in zebrafish (*Danio rerio*). In: Department of biology. Oslo: The University of Oslo; 2002, 104.
[150] Blanchard DC, Griebel G, Blanchard RJ. The mouse defense test battery: pharmacological and behavioral assays for anxiety and panic. Eur J Pharmacol 2003;463:97–116.
[151] Gerlai R, Ahmad F, Prajapati S. Differences in acute alcohol-induced behavioral responses among zebrafish populations. Alcohol Clin Exp Res 2008;32:1763–73.
[152] Richardson LF. The problem of contiguity. In: von Bertalanffy L, Rapoport A, editors. General systems yearbook of the society for general systems research. Society for General Systems Research; 1960. p. 139–187.
[153] Mandelbrot BB. How long is the coast of Britain? Statistical self-similarity and fractional dimension. Science 1967;155:636–8.

UNCORRECTED PROOF

Please cite this article in press as: Maximino C, et al. Measuring anxiety in zebrafish: A critical review. Behav Brain Res (2010), doi:10.1016/j.bbr.2010.05.031

[154] Rapp PE, Watanabe TAA, Faure P, Cellucci CJ. Nonlinear signal classification. Int J Bifurc Chaos Appl Sci Eng 2002;12:1273–93.
[155] Neumeister H, Cellucci CJ, Rapp PE, Korn H, Faber DS. Dynamical analysis reveals individuality of locomotion in goldfish. J Exp Biol 2004;207:697–708.
[156] Dielenberg RA, McGregor IS. Defensive behavior in rats towards predatory odors: a review. Neurosci Biobehav Rev 2001;25:597–609.
[157] Canteras NS. Neural systems activated in response to predators and partial predator stimuli. In: Blanchard RJ, Blanchard DC, Griebel R, Nutt D, editors. Handbook of anxiety and fear. Amsterdam: Elsevier B.V.; 2008. p. 125–39.
[158] Pitcher TJ. Who dares wins: the function and evolution of predator inspection behaviour in shoaling fish. Neth J Zool 1992;42:371–91.
[159] Dugatkin LA, Godin J. Prey approaching predators: a cost-benefit perspective. Ann Zool Fenn 1992;29:233–52.
[160] Magurran AE, Higham A. Information transfer across fish shoals under predator threat. Ethology 1988;78:153–8.
[161] Dugatkin LA. Tendency to inspect predators predict mortality risk in the guppy, *Poecilia reticulata*. Behav Ecol 1992;3:124–8.
[162] Yilmazer-Hanke DM, Roskoden T, Zilles K, Schwegler H. Anxiety-related behavior and densities of glutamate, $GABA_A$, acetylcholine and serotonin receptors in the amygdala of seven inbred mouse strains. Behav Brain Res 2003;145:145–59.
[163] Adamec RE, Shallow T. Lasting effects on rodent anxiety of a single exposure to a cat. Physiol Behav 1993;54:101–9.
[164] von Frisch K. Zur Psychologie des Fisch-Schwarmes. Naturwissenschaften 1938;26:601–6.
[165] Gerlai R, Csányi V. Genotype-environment interaction and the correlation structure of behavioral elements in paradise fish (*Macropodus opercularis*). Physiol Behav 1990;47:343–56.
Q4 [166] Gerlai R. Zebrafish antipredatory responses: a future for translational research? Behav Brain Res No prelo.
[167] Pfeiffer W. The distribution of fright reaction and alarm substance cells in fishes. Copeia 1977;4:653–65.
[168] Rehnberg BG, Schreck CB. Chemosensory detection of predators by coho salmon (*Oncorhynchus kisutch*): behavioral reaction and the physiological stress response. Can J Zool 1987;65:481–5.
[169] Brown GE, Adrian Jr JC, Smyth E, Leet H, Brennan S. Ostariophysan alarm substances: laboratory and field tests of the functional significance of nitrogen oxides. J Chem Ecol 2000;26:139–54.
[170] Kelly JM, Adrian Jr JC, Brown GE. Can the ratio of aromatic skeletons explain cross-species responses within evolutionary conserved Ostariophysan alarm cues? Testing the purine-ratio hypothesis. Chemoecology 2006;16:93–6.
[171] Suboski MD, Bain S, Carty AE, McQuoid LM, Seelen MI, Seifert M. Alarm reaction in acquisition and social transmission of simulated-predator recognition by zebra danio fish (*Brachydanio rerio*). J Comp Psychol 1990;104:101–12.
[172] Rehnberg BG, Bates EH, Smith RJF, Sloley BD, Richardson JS. Brain benzodiazepine receptors in fathead minnows and the behavioral response to alarm pheromone. Pharmacol Biochem Behav 1989;33:435–42.
[173] Prut L, Belzung C. The open field as a paradigm to measure the effects of drugs on anxiety-like behaviors: a review. Eur J Pharmacol 2003;463:3–33.
[174] Gerlai R. Zebra fish: an uncharted behavior genetic model. Behav Genet 2003;33:461–8.
[175] Aihart MJ, Lee DH, Wilson TD, Miller BE, Miller MN, Skalko RG. Movement disorders and neurochemical changes in zebrafish larvae after bath exposure to fluoxetine (PROZAC). Neurotoxicol Teratol 2007;29:652–64.
[176] Bencan Z, Levin ED. The role of α7 and α4β2 nicotinic receptors in the nicotine-induced anxiolytic effect in zebrafish. Physiol Behav 2008;95:408–12.
[177] Peitsaro N, Kaslin J, Anichtchik OV, Panula P. Modulation of the histaminergic system and behaviour by α-fluoromethylhistidine in zebrafish. J Neurochem 2003;86:432–41.
[178] Dadda M, Domenichini A, Piffer L, Argenton F, Bisazza A. Early differences in epithalamic left-right asymmetry influence lateralization and personality of adult zebrafish. Behav Brain Res 2009;206:208–15.
[179] Boehmler W, Carr T, Thisse C, Thisse B, Canfield VA, Levenson R. D4 dopamine receptor genes of zebrafish and effects of the antipsychotic clozapine on larval swimming behaviour. Genes Brain Behav 2007;6:155–66.
[180] MacPhail RC, Brooks J, Hunter DK, Padnos B, Irons TD, Padilla S. Locomotion in larval zebrafish: influence of time of day, lighting and ethanol. Neurotoxicology 2009;30:52–8.
[181] Bencan Z, Sledge D, Levin ED. Buspirone, chlordiazepoxide and diazepam effects in a zebrafish model of anxiety. Pharmacol Biochem Behav 2009;94:75–80.
[182] Sackerman J, Donegan JJ, Cunningham CS, Nguyen NN, Lawless K, Gould GG. Zebrafish behavior in novel environments: effects of acute exposure to anxiolytic compounds and choice of *Danio rerio* line. Int J Comp Psychol 2010;23:43–61.
[183] Krystal JH, Tabakoff B. Ethanol abuse, dependence, and withdrawal: neurobiology and clinical implications. In: Davis KL, Charney D, Coyle JT, Nemeroff C, editors. Neuropsychopharmacology: the fifth generation of progress. New York: Raven Press/American College of Neuropsychopharmacology; 2002.
[184] Clineschmidt BV. Effect of the benzodiazepine receptor antagonist Ro 15–1788 on the anticonvulsant and anticonflict actions of MK-801. Eur J Pharmacol 1982;84:119–21.
[185] Clineschmidt BV, Williams M, Witoslawski JJ, Bunting PR, Risley EA, Totaro JA. Restoration of shock-suppressed behavior by treatment with (+)-5-methyl-10,11-dihydro-5H-dibenzo[a, d]cyclohepten-5, 10-imine (MK-801), a substance with potent anticonvulsant, central sympathomimetic, and apparent anxiolytic properties. Drug Dev Res 1982;2:147–63.
[186] Blanchard DC, Blanchard RJ, Carobrez AdP, Veniegas R, Rodgers RJ, Shepherd JK. MK-801 produces a reduction in anxiety-related antipredator defensiveness in male and female rats and a gender-dependent increase in locomotor behavior. Psychopharmacology 1992;108:352–62.
[187] Davis RE, Klinger PD. NMDA receptor antagonist MK-801 blocks learning of conditioned stimulus-unconditioned stimulus contiguity but not fear of conditioned stimulus in goldfish (*Carassius auratus* L.). Behav Neurosci 1994;108:935–40.
[188] Kehne JH, McCloskey TC, Baron BM, Chi EM, Harrison BL, Whitten JP, et al. NMDA receptor complex antagonists have potential anxiolytic effects as measured with separation-induced ultrasonic vocalizations. Eur J Pharmacol 1991;193:283–92.
[189] Koek W, Colpaert FC. Use of a conflict procedure in pigeons to characterize anxiolytic drug activity: evaluation of *N*-methyl-D-aspartate antagonists. Life Sci 1991;49:PL37–42.
[190] Chen J, Patel R, Friedman TC, Jones KS. The behavioral and pharmacological actions of NMDA receptor antagonism are conserved in zebrafish larvae. Int J Comp Psychol 2010;23:82–90.
[191] Brown GE, Adrian Jr JC, Smyth E, Leet H, Brenna S. Ostariophysan alarm pheromones: laboratory and field tests of the functional significance of nitrogen oxides. J Chem Ecol 2000;26:139–54.
[192] El Yacoubi M, Ledent C, Parmentier M, Costentin J, Vaugeois J-M. The anxiogenic-like effect of caffeine in two experimental procedures measuring anxiety in the mouse is not shared by selective A_{2A} adenosine receptor antagonists. Psychopharmacology 2000;148:153–63.
[193] Schrijver NC, Bahr NI, Weiss IC, Wurbel H. Dissociable effects of isolation rearing and environmental enrichment on exploration, spatial learning and HPA activity in adult rats. Pharmacol Biochem Behav 2002;73:209–24.
[194] Brenes JC, Padilla M, Fornaguera J. A detailed analysis of open-field habituation and behavioral and neurochemical antidepressant-like effects in postweaning enriched rats. Behav Brain Res 2009;197:125–37.
[195] Espmark ÅM, Eriksen MS, Salte R, Braastad BO, Bakken M. A note on pre-spawning maternal cortisol exposure in farmed Atlantic salmon and its impact on the behaviour of offspring in response to a novel environment. Appl Anim Behav Sci 2008;110:404–9.
[196] Kofman O. The role of prenatal stress in the etiology of developmental behavioural disorders. Neurosci Biobehav Rev 2002;26:457–70.
[197] Leussis MP, Bolivar VJ. Habituation in rodents: a review of behavior, neurobiology, and genetics. Neurosci Biobehav Rev 2006;30:1045–64.
[198] Treit D, Menard J, Royan C. Anxiogenic stimuli in the elevated plus-maze. Pharmacol Biochem Behav 1993;44:463–9.
[199] Montgomery KC. The relation between fear induced by novel stimulation and exploratory behavior. J Comp Physiol Psychol 1955;48:254–60.
[200] Morato S, Brandão ML. Paradoxical increase of exploratory behaviour in the elevated plus maze by rats exposed to two kinds of aversive stimuli. Braz J Med Biol Res 1997;30:1113–20.
[201] Salum C, Roque-da-Silva AC, Morato S. Conflict as a determinant of rat behavior in three types of elevated plus-maze. Behav Proc 2003;63:87–93.
[202] File SE. Factors controlling measures of anxiety and responses to novelty in the mouse. Behav Brain Res 2001;125:151–7.
[203] Wright D, Nakamichi J, Krause J, Butlin RK. QTL analysis of behavioral and morphological differentiation between wild and laboratory zebrafish (*Danio rerio*). Behav Genet 2006;36:271–84.
[204] Lau B, Gould GG, Guo S. Bright light activates a zebrafish innate avoidance circuitry encompassing Dm, the teleostean homologue of pallial amygdala; 2010.
[205] Woodnorth MA, McNaughton N. Similar effects of medial supramammilary or systemic injections of chlordiazepoxide on both theta frequency and fixed-interval responding. Cogn Affect Behav Neurosci 2002;2:76–83.
[206] Chaouloff F, Durand M, Mormede P. Anxiety- and activity-related effects of diazepam and chlordiazepoxide in the rat light/dark and dark/light tests. Behav Brain Res 1997;85:27–35.
[207] Hascoet M, Bourin M. A new approach to the light/dark test procedure in mice. Pharmacol Biochem Behav 1998;60:645–53.
[208] Griebel G, Sanger DJ, Perrault G. Further evidence for differences between non-selective and BZ-1 (omega 1) selective, benzodiazepine receptor ligands in murine models of "state" and "trait" anxiety. Neuropharmacology 1996;35:1081–91.
[209] Shimada T, Matsumoto K, Osanai M, Matsuda H, Terasawa K, Watanabe H. The modified light/dark transition test in mice: evaluation of classic and putative anxiolytic and anxiogenic drugs. Gen Pharmacol 1995;26:205–10.
[210] Zangrossi Jr H, File SE. Chlordiazepoxide reduces the generalised anxiety, but not the direct responses, of rats exposed to cat odor. Pharmacol Biochem Behav 1992;43:1195–200.
[211] Ogwang SP. Inspection of a novel object by wild and laboratory zebrafish (*Danio rerio*). In: Department of biology. Bergen, Norway: University of Bergen; 2008. p. 68.
[212] Ennaceur A, Delacour J. A new one-trial test for neurobiological studies of memory in rats. 1: behavioural data. Behav Brain Res 1988;31:47–59.
[213] Dugatkin LA, Godin JG. Predator inspection, shoaling and foraging under predation hazard in the Trinidadian guppy, *Poecilia reticulata*. Environ Biol Fish 1992;34:265–76.
[214] Spence R, Gerlach G, Lawrence C, Smith C. The behaviour and ecology of the zebrafish, *Danio rerio*. Biol Rev 2008;83:13–34.

UNCORRECTED PROOF

Please cite this article in press as: Maximino C, et al. Measuring anxiety in zebrafish: A critical review. Behav Brain Res (2010), doi:10.1016/j.bbr.2010.05.031

[215] Engeszer RE, Patterson LB, Rao AA, Parichy DM. Zebrafish in the wild: a review of natural history and new notes from the field. Zebrafish 2007;4:21–40.
[216] Landeau L, Terborgh J. Oddity and the 'confusion effect' in predation. Anim Behav 1986;34:1372–80.
[217] van Olst JC, Hunter JR. Some aspects of the organization of fish shoals. J Fish Res Board Can 1970;27:1225–38.
[218] Lima SL. Back to the basic of anti-predatory vigilance: the group-size effect. Anim Behav 1995;49:11–20.
[219] Pitcher TJ, Parrish JK. Functions of shoaling behaviour in teleosts. In: Pitcher TJ, editor. Behaviour of teleost fishes. London: Chapman & Hall; 1993. p. 369–439.
[220] Ryer CH, Olla BL. Shifting the balance between foraging and predator avoidance: the importance of food distribution for a shoaling pelagic forager. Environ Biol Fish 1998;52:467–75.
[221] Becco C, Vandewalle N, Delcourt J, Poncin P. Experimental evidences of a structural and dynamical transition in fish school. Phys A 2006;367:487–93.
[222] Wright D. Repeated measures of shoaling tendency in zebrafish (*Danio rerio*) and other small teleost fishes. Nat Protoc 2006;1:1828–31.
[223] Partride BL, Pitcher TJ, Cullen JM, Wilson J. The three-dimensional structure of fish schools. Behav Ecol Sociobiol 1980;6:277–88.
[224] Fernandes Y, Gerlai R. Long-term behavioral changes in response to early developmental exposure to ethanol in zebrafish. Alcohol Clin Exp Res 2009;33:601–99.
[225] Tsubokawa T, Saito K, Kawano H, Kawamura K, Shinozuka K, Watanabe S. Pharmacological effects on mirror approaching behavior and neurochemical aspects of the telencephalon in the fish, medaka (*Oryzias latipes*). Social Neurosci 2009;4:276–86.
[226] Rex A, Voigt JP, Fink H. Anxiety but not arousal increases 5-hydroxytryptamine release in the rat ventral hippocampus *in vivo*. Eur J Neurosci 2005;22:1185–9.
[227] Barreto RE, Luchiari AC, Macondes AL. Ventilatory frequency indicates visual recognition of an allopatric predator in naïve Nile tilapia. Behav Proc 2003;60:235–9.
[228] Barreto RE, Volpato GL. Caution for using ventilatory frequency as an indicator of stress in fish. Behav Proc 2004;60:235–9.
[229] Brown C, Garnder C, Braithwaite V. Differential stress response in fish from areas of high- and low-predation pressure. J Comp Physiol B 2005;175:305–12.
[230] Rooney DJ, Laming PR. Localization of telencephalic regions concerned with habituation of cardiac and ventilatory responses associated with arousal in the goldfish (*Carassius auratus*). Behav Neurosci 1986;100:45–50.
[231] Canavello PR, Cachat JM, Beeson E, Laffoon AL, Grimes C, Haymore W, et al. Measuring endocrine (cortisol) responses of zebrafish to stress. In: Kalueff AV, Canavello PR, Cachat J, editors. Zebrafish neurobehavioral protocols. New York: Humana Press; in press.
[232] Goldsmith P. Zebrafish as a pharmacological tool: the how, why and when. Curr Op Pharmacol 2004;4:504–12.
[233] Dugatkin LA, McCall MA, Gregg RG, Cavanaugh A, Christensen C, Unseld M. Zebrafish (*Danio rerio*) exhibit individual differences in risk-taking behavior during predator inspection. Ethol Ecol Evol 2005;17:77–81.

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